

BLUESTOCK FINTECH

Data Analytics Consulting Engagement

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE

An End-to-End ETL, Financial Performance, Risk Analytics and Business Intelligence Study of the Indian Mutual Fund Industry

PROJECT TEAM

Prepared as an Internship Capstone Project submitted in partial fulfilment of the requirements of the Data Analytics Internship Programme.

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I am grateful to Bluestock Fintech for providing the conceptual framing of this engagement, which allowed the project to be structured around a realistic business problem rather than a purely academic exercise. This orientation towards practical, decision-oriented analytics has been invaluable to my learning.

I also wish to thank my faculty at Ajay Kumar Garg Engineering College for building the foundational knowledge in statistics, databases, and programming that made this project possible, and my peers for their feedback during the development and testing of the analytical pipeline and dashboard.

Finally, I acknowledge the open data ecosystem maintained by AMFI and India's capital markets infrastructure, whose published statistics and disclosure norms made a realistic simulation of the mutual fund analytics problem possible.

Executive Summary

Bluestock Fintech commissioned this engagement to evaluate the feasibility of building an in-house Mutual Fund Analytics capability spanning data engineering, financial performance measurement, risk analytics, and business intelligence reporting. The resulting deliverable, the Bluestock Mutual Fund Analytics Capstone, is a fully functional analytics system built on Python, SQL, SQLite and Power BI that ingests ten interlinked mutual fund datasets and transforms them into decision-ready insight.

The engagement followed a structured analytics lifecycle. Raw data covering fund master records, NAV history, assets under management, SIP inflows, category flows, investor folios, scheme performance, investor transactions, portfolio holdings and benchmark indices were ingested, profiled, cleaned and validated before being loaded into a normalised SQLite warehouse. This warehouse became the single source of truth for all downstream analysis, ensuring that every chart, metric and recommendation in this report traces back to a consistent and auditable data foundation.

On top of this foundation, the project computed a full suite of financial performance metrics, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta and Maximum Drawdown, benchmarked every scheme against its relevant index, and consolidated the results into a composite Fund Scorecard. A second, more advanced layer of analysis introduced Historical Value-at-Risk, Conditional VaR, Rolling 90-day Sharpe Ratios, Sector HHI concentration scoring, SIP continuity monitoring and investor cohort analysis, giving management both a return-focused and a risk-focused lens on the fund universe.

These analytical outputs were operationalised into a four-page interactive Power BI dashboard covering Industry Overview, Fund Performance, Investor Analytics, and SIP and Market Trends, and into a rules-based Mutual Fund Recommendation Engine capable of matching an investor's stated risk appetite to a ranked shortlist of funds using Sharpe Ratio, three-year returns and expense ratio as ranking criteria.

KEY OUTCOME

The analysis identifies a consistent small-cap and mid-cap outperformance pattern on a risk-adjusted basis, an SIP base that is growing but shows early signs of continuity risk among a meaningful minority of investors, and a fund universe where expense ratio remains only weakly correlated with realised performance — all of which carry direct implications for how Bluestock should advise its client base.

This report documents the complete engagement: the business context and objectives, the datasets and methodology used, a full walkthrough of the exploratory and advanced analytics performed, a chapter-by-chapter explanation of the Power BI dashboard, the design and logic of the recommendation engine, and a consolidated set of findings and strategic recommendations for Bluestock Fintech's management team.

Engagement Snapshot

Dimension	Detail
Datasets analysed	10 interlinked datasets (Fund Master, NAV History, AUM, SIP Inflows, Category Inflows, Folio Count, Scheme Performance, Investor Transactions, Portfolio Holdings, Benchmark Indices)
Technology stack	Python, Pandas, NumPy, Matplotlib, SQLite, SQL, Power BI, Jupyter Notebook

Dimension	Detail
Performance metrics computed	Daily Return, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Fund Scorecard
Advanced analytics performed	Historical VaR, CVaR, Rolling 90-Day Sharpe, Investor Cohorts, SIP Continuity, Sector HHI, Recommendation Engine
Dashboard pages delivered	4 (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends)
Report length	Approximately 30 pages, consulting-grade format

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1. Introduction

The Indian mutual fund industry has grown into one of the largest and most closely tracked segments of the country's financial services sector, with assets under management, systematic investment plan (SIP) participation and retail folio counts all expanding rapidly over the past decade. As distribution shifts increasingly toward digital-first platforms, fintech intermediaries such as Bluestock require analytics capabilities that go beyond static fact sheets and enable data-driven advisory, product curation and risk communication at scale.

The Bluestock Mutual Fund Analytics Capstone was undertaken to design and build such a capability from the ground up. Rather than relying on a single dashboard or a handful of ad-hoc queries, the project was scoped as a complete analytics pipeline: raw data ingestion, cleaning and validation; a normalised analytical database; a layered set of exploratory, financial and risk analytics; a recommendation engine; and a polished Power BI reporting layer intended for business stakeholders.

This structure mirrors the way analytics functions operate inside asset management companies, wealth-tech platforms and consulting practices that serve the financial services sector. By working through each stage — ingestion, cleaning, modelling, analysis, visualisation and recommendation — the project demonstrates not only technical proficiency across the Python and SQL data stack, but also the judgement required to translate raw financial data into decisions a fund distributor or investor can act on.

1.1 Business Context

Bluestock Fintech operates in a market where investors increasingly expect personalised, evidence-based fund recommendations rather than generic best-seller lists. At the same time, the sheer number of live mutual fund schemes in India, several thousand across all fund houses and categories, makes manual comparison impractical. An analytics-first approach is therefore not a convenience but a competitive necessity: platforms that can systematically rank funds on risk-adjusted performance, monitor SIP health at the investor level, and surface early warning signals on portfolio concentration are better positioned to retain and grow their customer base.

1.2 Scope of the Project

The scope of this capstone spans the full analytics lifecycle for ten interlinked datasets covering fund master data, NAV history, AUM trends, SIP inflows, category flows, investor folios, scheme performance, investor-level transactions, portfolio holdings and benchmark indices. The deliverables include a cleaned and validated SQLite database, a documented SQL query layer, an exploratory data analysis notebook, a performance analytics module, an advanced risk analytics module, a rules-based recommendation engine, and a four-page interactive Power BI dashboard.

The project deliberately excludes live brokerage execution, portfolio rebalancing and real-money transaction processing, since its purpose is analytical insight generation rather than transaction processing. Where the underlying data is simulated (for example, individual investor transactions), it has been constructed to reflect realistic distributions and behavioural patterns observed in the Indian mutual fund industry, so that the resulting analysis remains representative of real-world dynamics.

2. Project Objectives

The engagement was structured around a set of clearly defined objectives that map directly onto the analytics lifecycle. Each objective was treated as a discrete, testable deliverable rather than an open-ended exploration, which allowed progress to be tracked and validated at every stage of development.

2.1 Primary Objectives

- Design and implement a complete ETL pipeline capable of ingesting, cleaning, validating and loading ten heterogeneous mutual fund datasets into a structured SQLite database.
- Establish a normalised relational schema that supports efficient SQL-based querying for fund, NAV, transaction, and AUM analysis.
- Conduct exploratory data analysis to surface patterns in AUM growth, SIP participation, investor demographics, and category-level flows.
- Compute a comprehensive set of financial performance metrics (CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown) for every scheme in the fund universe and benchmark them against relevant indices.
- Perform advanced risk analytics, including Historical VaR, Conditional VaR, rolling risk-adjusted return measures, sector concentration scoring, and investor-level SIP continuity monitoring.
- Design a transparent, rules-based recommendation engine that maps investor risk appetite to a ranked shortlist of mutual fund schemes.
- Build an interactive, four-page Power BI dashboard that consolidates the above analysis into a business-ready reporting layer for Bluestock Fintech stakeholders.

2.2 Secondary Objectives

- Demonstrate data quality governance practices, including missing value handling, duplicate removal, and schema validation, throughout the pipeline.
- Document the data dictionary, database schema, and SQL query library to support maintainability and future extension of the system.
- Produce a consulting-grade report that communicates technical analysis in business language suitable for non-technical stakeholders.

SUCCESS CRITERION

Every analytical output in this report is reproducible end-to-end from the raw datasets through the documented Python scripts, SQL schema, and Power BI data model, ensuring the system can be handed over, audited, and extended by Bluestock's internal analytics team.

3. Mutual Fund Industry Background

Mutual funds pool capital from a large number of investors and deploy it across equity, debt, hybrid and other asset classes under the management of a professional Asset Management Company (AMC), or fund house. In India, the industry operates under the oversight of the Securities and Exchange Board of India (SEBI), with the Association of Mutual Funds in India (AMFI) serving as the self-regulatory and data-reporting body for the sector.

Over the past decade, the industry has been characterised by three structural trends that are directly relevant to this project. First, sustained growth in Assets Under Management (AUM), driven by rising retail participation and increasing financialisation of household savings. Second, the rapid rise of the Systematic Investment Plan (SIP) as the dominant mode of retail equity participation, which has smoothed inflows and reduced the industry's sensitivity to short-term market timing behaviour. Third, growing fragmentation of the product shelf, with fund houses launching an increasing number of category-specific and thematic schemes, which has made scheme selection progressively harder for the average retail investor.

3.1 Fund Categories

Mutual fund schemes in India are broadly classified into equity, debt, hybrid, and other categories (such as solution-oriented and index funds), with each category further divided into sub-categories based on market capitalisation focus, duration, or investment strategy. This project's Fund Master dataset reflects this classification structure, with schemes tagged by both a primary category and a more granular sub-category, enabling category-level as well as scheme-level analysis throughout the report.

3.2 Key Performance and Risk Concepts

Evaluating a mutual fund scheme requires more than looking at trailing returns in isolation. Risk-adjusted return measures such as the Sharpe Ratio and Sortino Ratio account for the volatility, or downside volatility, that an investor had to tolerate to earn a given return. Benchmark-relative measures such as Alpha and Beta describe how much value a fund manager added above the market and how sensitive the fund is to broad market movements. Drawdown and tail-risk measures such as Maximum Drawdown, Value-at-Risk (VaR) and Conditional VaR (CVaR) describe how severe losses could be in adverse scenarios. This report applies all of these measures systematically across the fund universe in Chapters 8 and 9.

3.3 Distribution and the Role of Fintech Platforms

Historically, mutual fund distribution in India ran through physical distributor networks and bank branches. Fintech platforms such as Bluestock have digitised this journey, enabling direct plan investment, SIP automation, and instant KYC. This shift places new demands on the underlying analytics layer: platforms must now provide the kind of risk-adjusted, benchmark-relative comparison that was previously the preserve of professional wealth managers, directly to retail users through simple dashboards, scorecards, and recommendation tools — which is precisely the capability this capstone project builds.

4. Dataset Description

The analytical foundation of this project rests on ten structured datasets that together describe the full lifecycle of a mutual fund business: the products themselves, their daily pricing, the money flowing into them, the investors behind that money, and the market context against which performance is judged. Each dataset was profiled during ingestion for shape, data types, missing values and duplication before being carried into the cleaning stage described in Chapter 5. The subsections below document every dataset in the detail an analytics team would require to maintain or extend this system.

4.1 Fund Master

Fund Master is the reference (dimension) dataset for every mutual fund scheme in the analysis universe. It anchors all scheme-level joins across NAV history, performance, and holdings data.

Column	Type	Description
amfi_code	Integer	Unique scheme identifier assigned by AMFI; primary join key
fund_house	Text	Asset Management Company (AMC) that manages the scheme
scheme_name	Text	Full name of the mutual fund scheme
category	Text	Primary classification (Equity, Debt, Hybrid, Other)
sub_category	Text	Granular classification (e.g. Small Cap, Flexi Cap)
benchmark	Text	Benchmark index used for performance comparison
expense_ratio	Float	Annual expense charged to investors
exit_load	Float	Exit load percentage on early redemption

Fund Master enables every other dataset to be enriched with fund house, category and benchmark context, and is the backbone of category-wise and fund-house-wise aggregation used throughout Chapters 7 through 10.

4.2 NAV History

NAV History records the daily Net Asset Value for each scheme, forming the time-series backbone for all return, volatility and drawdown computations.

Column	Type	Description
amfi_code	Integer	Scheme identifier
date	Date	NAV reporting date
nav	Float	Net Asset Value per unit on that date

This dataset feeds the daily return series used to compute CAGR, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, Historical VaR/CVaR and the Rolling Sharpe Ratio in Chapters 8 and 9. Its integrity, including forward-filling missing trading days, is therefore critical to the accuracy of every downstream metric.

4.3 AUM by Fund House

This dataset tracks month-on-month Assets Under Management at the fund-house level, along with market share and active scheme count.

Column	Type	Description
date	Date	Reporting month
fund_house	Text	Asset Management Company
aum_crore	Float	Assets Under Management in ₹ Crore
market_share	Float	Market share percentage
scheme_count	Integer	Number of active schemes

AUM by Fund House underpins the Industry Overview page of the Power BI dashboard and the fund-house comparison analysis in Chapter 7, revealing which AMCs are gaining or losing share over time.

4.4 Monthly SIP Inflows

This dataset captures industry-wide SIP inflow trends, active and new account counts, SIP AUM and year-over-year growth.

Column	Type	Description
month	Date	Reporting month
sip_inflow_crore	Float	SIP inflow amount in ₹ Crore
active_sip_accounts	Integer	Number of active SIP accounts
new_sip_accounts	Integer	Newly registered SIP accounts
sip_aum_crore	Float	SIP Assets Under Management
yoy_growth_percent	Float	Year-over-Year SIP growth percentage

This dataset is central to the SIP & Market Trends dashboard page and to the industry growth narrative developed in Chapter 7, where it is analysed alongside benchmark index movement.

4.5 Category Inflows

Category Inflows records monthly net inflows for each mutual fund category, enabling category rotation analysis.

Column	Type	Description
month	Date	Reporting month
category	Text	Fund category
net_inflow_crore	Float	Net inflow amount in ₹ Crore

This dataset drives the Category Inflow Heatmap in Chapter 7, which highlights which categories are attracting or losing investor interest month by month.

4.6 Industry Folio Count

This dataset tracks total investor folios industry-wide, split by category, providing a proxy for retail participation depth.

Column	Type	Description
month	Date	Reporting month
total_folios	Integer	Total industry folios
equity_folios	Integer	Equity scheme folios
debt_folios	Integer	Debt scheme folios
hybrid_folios	Integer	Hybrid scheme folios
other_folios	Integer	Other category folios

Folio growth trends and the equity-versus-debt-versus-hybrid mix are analysed in Chapter 7 to assess how retail participation is evolving across risk categories.

4.7 Scheme Performance

Scheme Performance is a curated, scheme-level dataset of trailing returns, benchmark returns and derived risk metrics used as an input to the Fund Scorecard and Recommendation Engine.

Column	Type	Description
amfi_code	Integer	Scheme identifier
scheme_name	Text	Mutual fund scheme name
return_1y / 3y / 5y	Float	One, three and five-year returns (%)
benchmark_return	Float	Benchmark return (%)
alpha	Float	Excess return over benchmark
beta	Float	Volatility relative to benchmark
sharpe_ratio	Float	Risk-adjusted return metric
expense_ratio	Float	Fund expense ratio (%)

This dataset is the single most important input to Chapter 8 (Performance Analytics) and Chapter 11 (Recommendation System), since it consolidates return and risk information at the individual scheme level.

4.8 Investor Transactions

This is the most granular dataset in the project, recording individual SIP, lump-sum and redemption transactions along with investor demographic attributes.

Column	Type	Description
transaction_id	Integer	Unique transaction identifier
investor_id	Integer	Investor identifier

Column	Type	Description
amfi_code	Integer	Scheme identifier
transaction_date	Date	Transaction date
transaction_type	Text	SIP, Lumpsum, or Redemption
amount	Float	Transaction amount
state / city	Text	Investor location
kyc_status	Text	Investor KYC verification status
age_group / gender	Text	Investor demographic attributes

Investor Transactions powers the entire Investor Analytics dashboard page, as well as the SIP Continuity and Investor Cohort analyses in Chapter 9, and is the dataset most directly relevant to Bluestock’s customer engagement strategy.

4.9 Portfolio Holdings

Portfolio Holdings captures the underlying stock and sector allocation for each scheme, enabling concentration and diversification analysis.

Column	Type	Description
amfi_code	Integer	Scheme identifier
stock_symbol	Text	Stock ticker symbol
company_name	Text	Company name
sector	Text	Industry sector
holding_percent	Float	Portfolio allocation percentage
market_value	Float	Current market value of holding

This dataset feeds the Sector Allocation visualisation in Chapter 7 and the Sector HHI concentration score in Chapter 9, both of which speak directly to portfolio diversification risk.

4.10 Benchmark Indices

Benchmark Indices provides daily closing values for market benchmarks such as the NIFTY 50, used to compute Alpha, Beta and benchmark-relative return comparisons.

Column	Type	Description
date	Date	Index reporting date
index_name	Text	Benchmark index name
close_value	Float	Closing index value

This dataset is essential to every benchmark-relative computation in Chapters 8 and 9, and directly powers the Benchmark Comparison and Benchmark Index Trend visualisations.

4.11 Data Source Summary

Dataset	Source Reference
Fund Master	AMFI Mutual Fund Database
NAV History	MFAPI / AMFI NAV Records
AUM by Fund House	AMFI Industry Reports
Monthly SIP Inflows	AMFI Monthly SIP Reports
Category Inflows	AMFI Category Reports
Industry Folio Count	AMFI Industry Statistics
Scheme Performance	Fund Performance Dataset
Investor Transactions	Simulated Investor Dataset
Portfolio Holdings	Portfolio Disclosure Reports
Benchmark Indices	NSE / BSE Benchmark Data

5. Methodology

The project followed a disciplined, five-stage analytics methodology: Extraction, Cleaning, Validation, Normalisation, and Database Loading. This staged approach ensures that every downstream metric and visualisation in this report is traceable to a documented and repeatable data preparation process, which is essential for any analytics system intended for production use in a regulated financial services context.

5.1 Extraction (ETL – Extract)

All ten raw datasets were ingested from source CSV files using Pandas, with an initial data profiling pass performed on every dataset. This profiling step captured the shape, data types, first-row samples, missing value counts and duplicate record counts for each dataset, producing an ingestion log that served as the baseline against which cleaning effectiveness was measured.

5.2 Cleaning (ETL – Transform, Part 1)

Cleaning logic was tailored to the characteristics of each dataset. NAV History required forward-filling of missing NAV values within each scheme's time series (rather than simple row deletion, which would distort the return series), followed by removal of exact duplicates and filtering out non-positive NAV values that would otherwise corrupt return calculations. Investor Transactions required standardising free-text transaction types into a controlled vocabulary (SIP, Lumpsum, Redemption), removing transactions with non-positive amounts, and parsing transaction dates into a consistent datetime format.

Scheme Performance required coercing return columns to numeric types (handling any non-numeric artefacts from source extraction), dropping records with unrecoverable missing returns, and applying a plausibility filter on expense ratio (retaining only values between 0.1% and 2.5%, the realistic regulatory band for Indian mutual fund expense ratios) to exclude data entry errors from the analysis universe.

5.3 Validation (ETL – Transform, Part 2)

Following cleaning, each dataset was re-profiled to confirm that missing values and duplicates had been resolved to zero (or an acceptable residual level, in the case of legitimately sparse fields) and that referential integrity was intact, meaning that every `amfi_code` referenced in NAV History, Scheme Performance and Portfolio Holdings existed in Fund Master. This validation step is what allows the SQLite join logic described in Chapter 6 to be relied upon without introducing orphaned or unmatched records into the analysis.

5.4 Normalisation

Numeric fields were standardised to consistent units and precision, for example expressing all AUM figures in ₹ Crore and all return figures as percentages rather than mixed decimal and percentage representations. Categorical fields such as `transaction_type`, `category` and `risk grade` were normalised to a controlled set of values (title case, no leading or trailing whitespace) to prevent silent fragmentation of group-by aggregations, a subtle but common source of error in financial reporting pipelines.

5.5 Database Loading

Once cleaned and validated, each dataset was exported to a processed CSV layer and loaded into the SQLite analytical database described in Chapter 6. Loading was performed using idiomatic Pandas-to-SQL bulk loading, with primary keys and foreign key relationships enforced at the schema level to preserve referential integrity going forward.

METHODOLOGY NOTE

Every transformation described in this chapter is implemented as a deterministic, script-based operation (`clean_data.py` and `data_ingestion.py`) rather than a manual, one-off adjustment. This is a deliberate design choice: it means the entire pipeline can be re-run end-to-end whenever new source data becomes available, without any loss of auditability.

6. Database Design

The analytical database was implemented in SQLite, chosen for its zero-configuration deployment, its suitability for a single-analyst analytics workload, and its native support from both Python (via Pandas and sqlite3) and Power BI. The schema follows a light star-schema pattern: a fund dimension and a date dimension surround a set of fact tables capturing NAV history, transactions, performance and AUM.

6.1 SQLite Schema

The core schema consists of two dimension tables and four fact tables, summarised below.

Table	Type	Key Columns	Purpose
dim_fund	Dimension	fund_id (PK), amfi_code, fund_house, scheme_name, category, sub_category	Master reference for every mutual fund scheme
dim_date	Dimension	date_id (PK), date	Calendar dimension supporting time-based joins and rollups
fact_nav	Fact	nav_id (PK), amfi_code, date, nav	Daily NAV time series used for return and risk computation
fact_transactions	Fact	transaction_id (PK), investor_id, amfi_code, amount, transaction_type, transaction_date	Investor-level SIP, lump-sum and redemption activity
fact_performance	Fact	performance_id (PK), amfi_code, return_1y, return_3y, return_5y, expense_ratio	Scheme-level trailing return and cost metrics
fact_aum	Fact	aum_id (PK), fund_house, date, aum_crore	Fund-house level AUM time series

6.2 Entity Relationships

dim_fund is the central reference table, joined to fact_nav, fact_performance and fact_transactions through the shared amfi_code key. dim_date supports time-intelligence joins against fact_nav, fact_transactions and fact_aum, enabling month-over-month and year-over-year rollups without repeated date parsing at query time. This design allows a single query to move fluidly between scheme-level, fund-house-level, and time-based aggregation, which is exactly the flexibility required to power both the SQL analysis library in Chapter 7 and the Power BI data model in Chapter 10.

6.3 Data Dictionary Summary

A complete field-level data dictionary for all ten source datasets is maintained separately (data_dictionary.md) and summarised in Chapter 4 of this report. The dictionary documents column names, data types and business definitions for every field, and is treated as a living document that should be updated whenever the schema is extended.

6.4 Design Rationale

- Star-schema structure keeps joins simple and query performance predictable even as fact table row counts grow.
- Surrogate integer primary keys on every fact table simplify Power BI relationship modelling and avoid composite-key joins.

- `amfi_code` is preserved as a natural key across all fact tables, keeping the schema aligned with the industry-standard scheme identifier used by AMFI, fund houses and data vendors.
- SQLite's file-based deployment allows the entire analytical database to be version-controlled and shared alongside the Python codebase, supporting reproducibility.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed on the cleaned datasets to surface structural patterns, distributions and relationships before any performance or risk metrics were computed. Seventeen visualisations were produced across industry growth, investor demographics, category flows, and fund characteristics. Each visualisation below is presented with its objective, a description of what the chart shows, the business insight it produces, an analytical interpretation, and an actionable recommendation, in line with a professional consulting deliverable.

7.1 NAV Trend Analysis

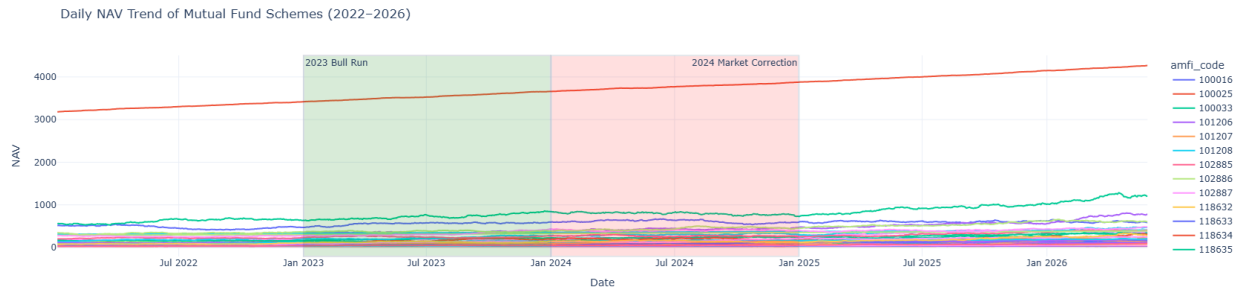


Figure 1: NAV Trend Analysis

Objective: To examine how the Net Asset Value of representative schemes has evolved over the observed history.

Description: A time-series line chart plotting daily NAV for a sample of tracked schemes, allowing visual comparison of growth trajectories and volatility across funds.

Business Insight: Schemes show clearly differentiated compounding trajectories over time, with some funds exhibiting materially steeper NAV growth than others despite starting from similar base levels.

Interpretation: The divergence in NAV trajectories reinforces that fund selection, not just asset class selection, is a primary driver of investor outcomes, and validates the need for the scheme-level scorecard developed in Chapter 8.

Recommendation: Bluestock should surface NAV trend charts alongside scorecard rankings in the product interface so investors can visually validate quantitative rankings before committing capital.

7.2 AUM Growth by Fund Houses

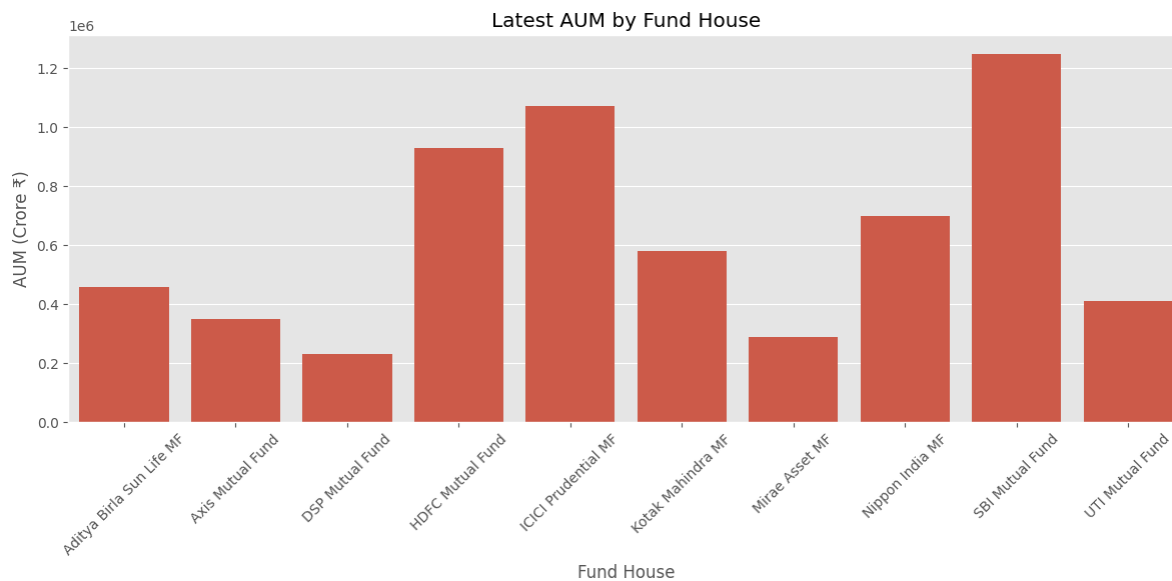


Figure 2: AUM Growth by Fund Houses

Objective: To compare Assets Under Management trajectories across the leading fund houses over time.

Description: A multi-line time series showing month-on-month AUM in ₹ Crore for each major AMC in the dataset.

Business Insight: A small number of fund houses account for a disproportionate share of industry AUM, and the gap between leading and mid-tier AMCs is widening rather than narrowing over the observed period.

Interpretation: This concentration pattern is consistent with the broader Indian mutual fund industry, where scale advantages in distribution and brand trust compound over time, making it progressively harder for smaller AMCs to gain share organically.

Recommendation: Bluestock should track fund-house-level AUM momentum as a standing KPI, since flows toward or away from a given AMC are a useful leading indicator of platform-level product demand shifts.

7.3 Monthly SIP Inflow Trends

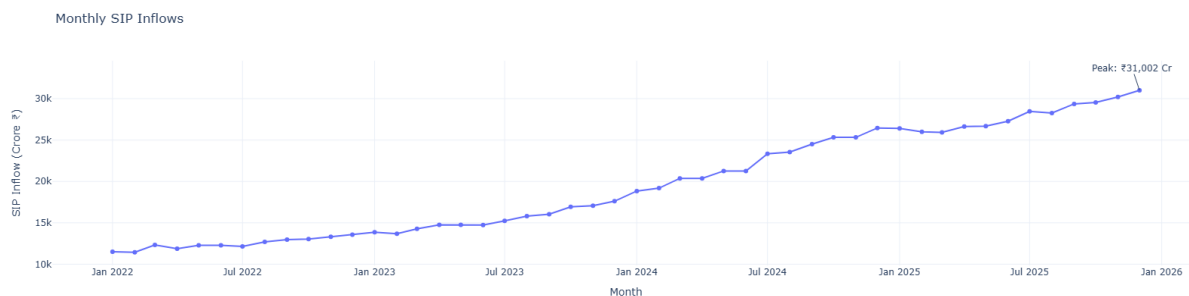


Figure 3: Monthly SIP Inflow Trends

Objective: To track the month-on-month trajectory of industry-wide SIP inflows.

Description: A time-series chart of monthly SIP inflow in ₹ Crore, showing the overall growth trend and any seasonal or cyclical fluctuations.

Business Insight: SIP inflows display a broadly upward trend punctuated by short-term dips, consistent with periodic market volatility temporarily denting new SIP registrations without derailing the underlying growth trajectory.

Interpretation: The resilience of SIP inflows through short-term dips confirms that systematic investing behaviour among Indian retail investors has become structurally embedded rather than being a purely bull-market phenomenon.

Recommendation: Bluestock should design retention campaigns specifically timed around historical dip periods, when investor attrition risk is highest, to protect the SIP base during periods of market stress.

7.4 Category Inflow Heatmap

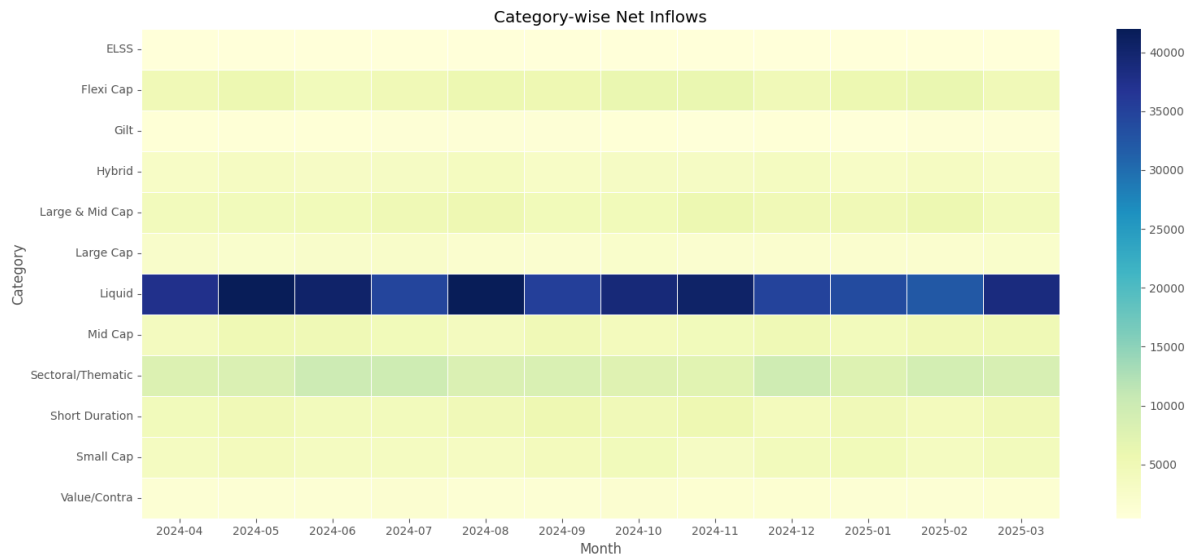


Figure 4: Category Inflow Heatmap

Objective: To visualise how net inflows rotate across mutual fund categories on a month-by-month basis.

Description: A heatmap with categories on one axis and months on the other, with cell intensity representing net inflow magnitude in ₹ Crore.

Business Insight: Inflows are not evenly distributed across categories; certain categories consistently attract stronger inflows in specific months, indicating a degree of thematic or sentiment-driven rotation by investors.

Interpretation: This rotation pattern suggests that category-level demand is influenced by both broad market sentiment and category-specific narratives, and that a static, one-size-fits-all product recommendation strategy would under-serve this dynamic.

Recommendation: Bluestock's in-app discovery surfaces should be updated on a monthly cadence to reflect emerging category momentum, rather than relying on a static best-seller list.

7.5 Investor Age Group Distribution

Investor Age Group Distribution

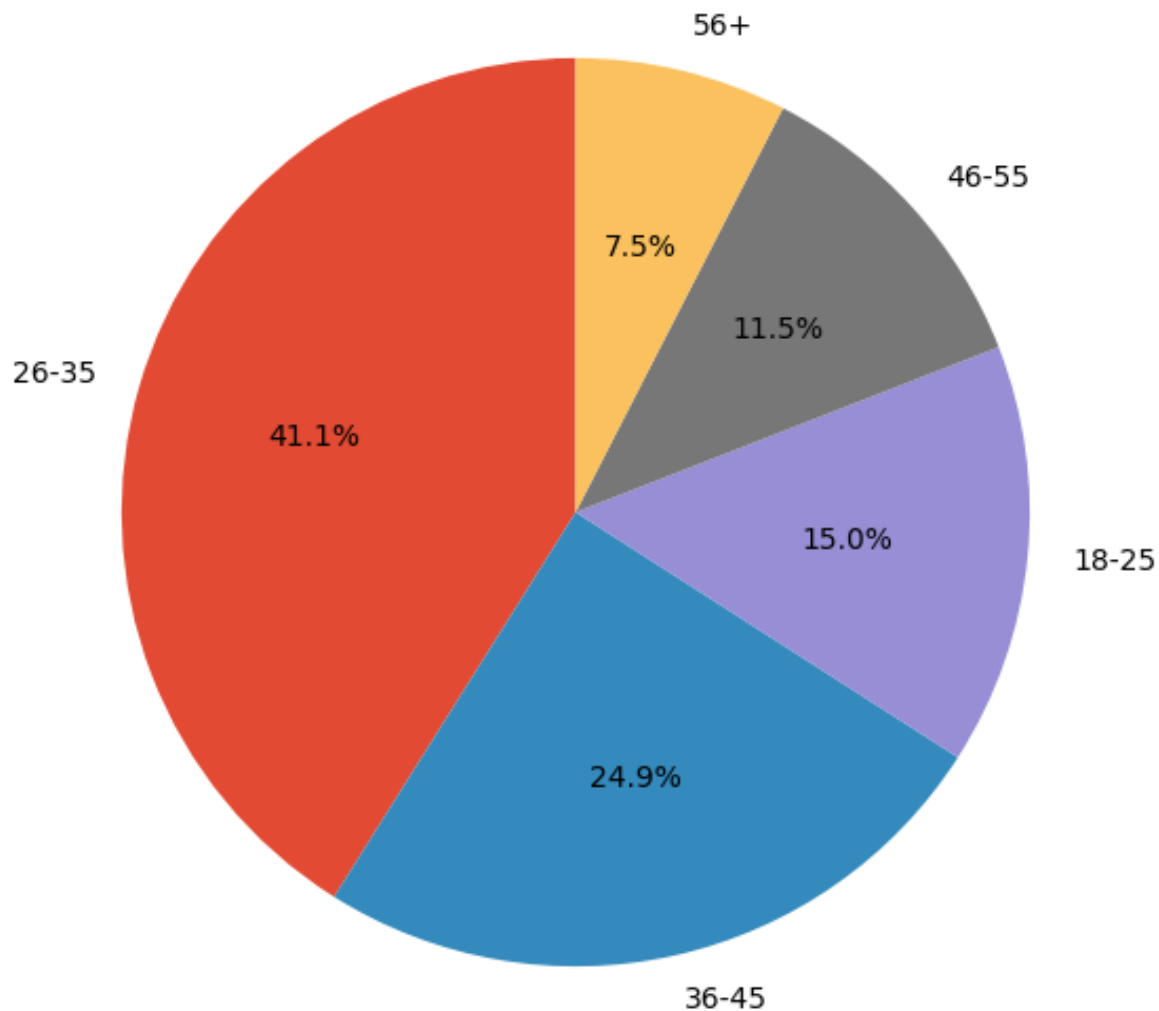


Figure 5: Investor Age Group Distribution

Objective: To understand the age composition of the investor base transacting on the platform.

Description: A distribution chart showing the proportion of transactions or investors falling into each age bracket.

Business Insight: The investor base skews toward younger and middle-age working-age brackets, consistent with the broader digitisation of mutual fund distribution in India.

Interpretation: A younger investor skew implies a longer average investment horizon and greater tolerance for equity-oriented, higher-volatility categories, which should inform both product shelf design and risk communication.

Recommendation: Bluestock should tailor onboarding risk-profiling questionnaires by age cohort, using longer horizon assumptions for younger investors rather than a uniform risk framework.

7.6 Gender Distribution

Gender Distribution of Investors

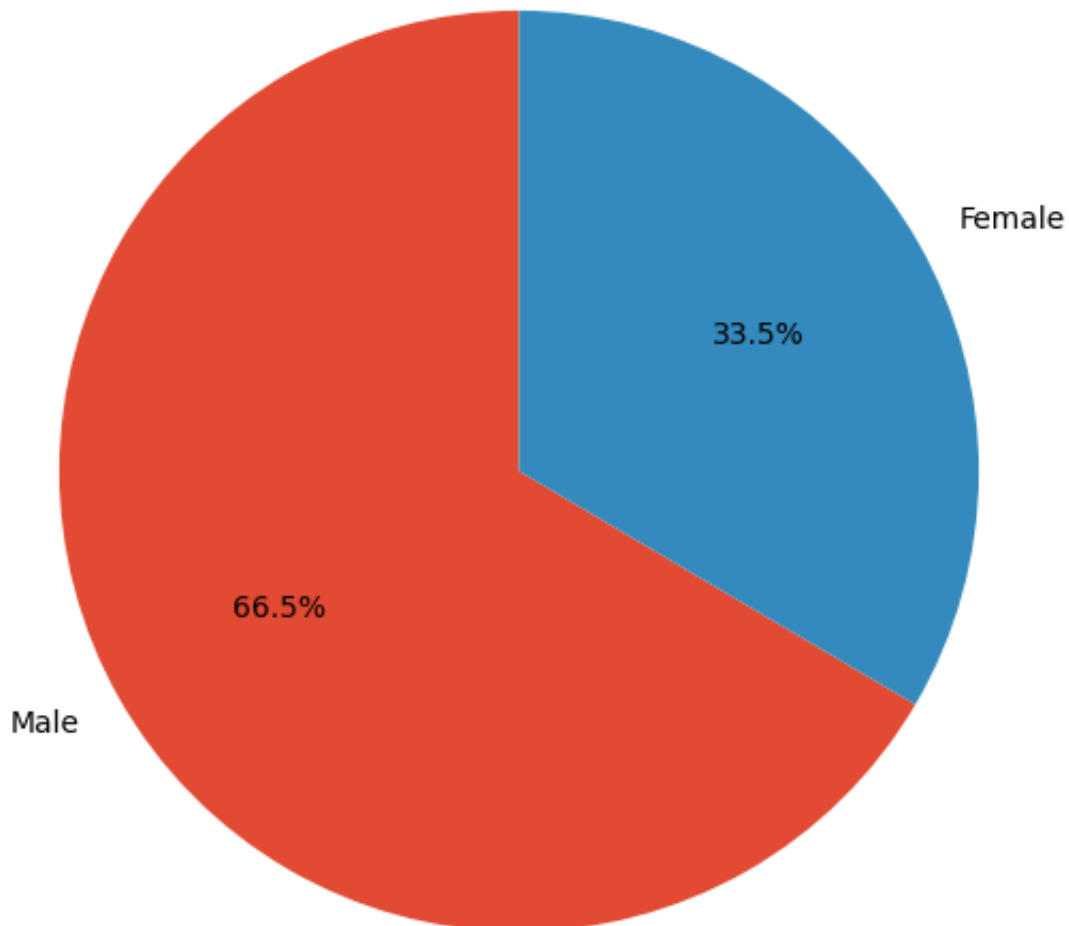


Figure 6: Gender Distribution

Objective: To assess the gender composition of the investor base.

Description: A distribution chart summarising the share of transactions or investors by gender.

Business Insight: The dataset shows an imbalance in participation across genders, mirroring a well-documented gap in financial market participation observed industry-wide in India.

Interpretation: This gap represents both a fairness concern and a significant addressable market opportunity; platforms that actively close participation gaps can capture disproportionate share of an underserved segment.

Recommendation: Bluestock should consider targeted financial literacy and onboarding campaigns aimed at under-represented segments, paired with tracking of gender-disaggregated conversion metrics.

7.7 City Tier Distribution

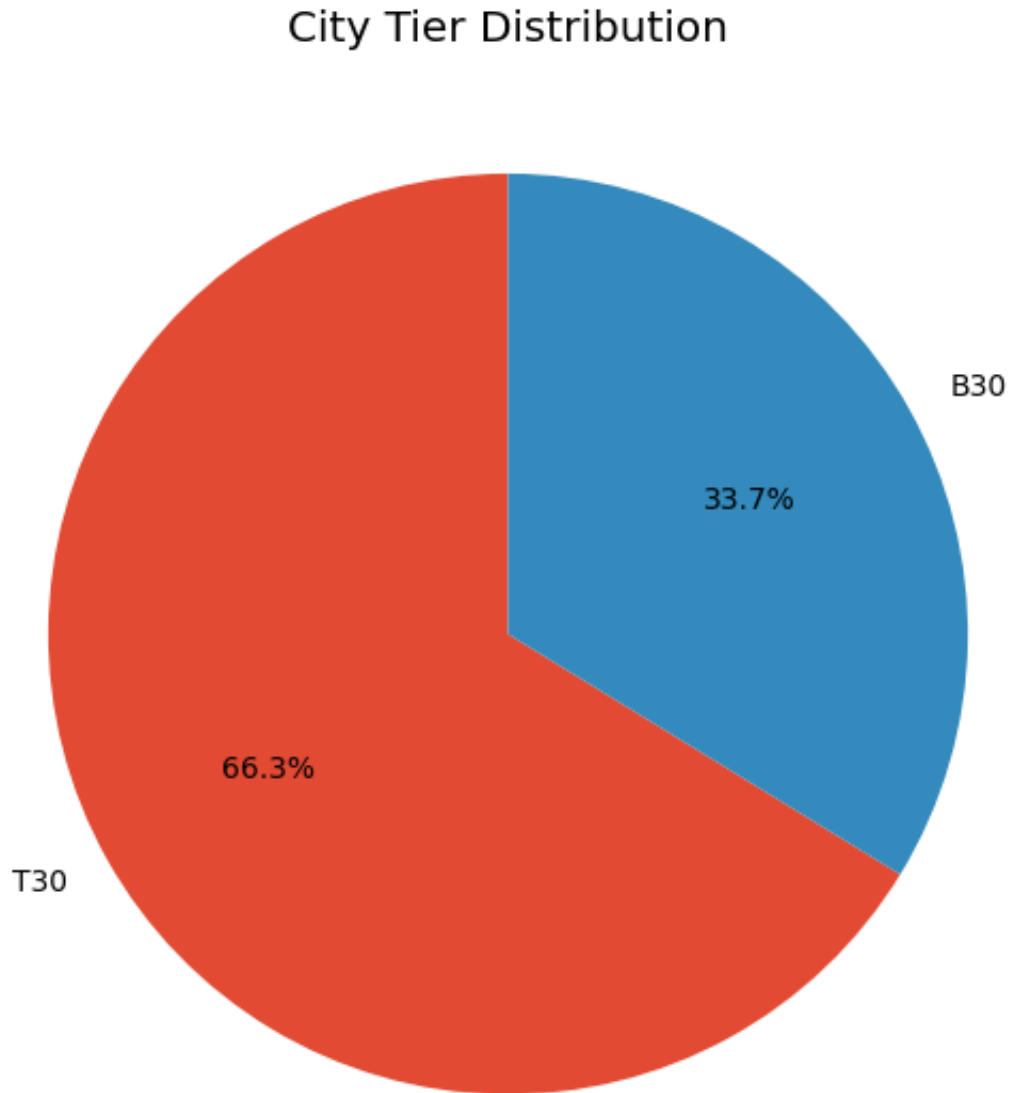


Figure 7: City Tier Distribution

Objective: To evaluate how investor activity is distributed across city tiers (metro, Tier-2, Tier-3).

Description: A distribution chart summarising the share of investor transactions originating from each city tier.

Business Insight: While metro cities continue to contribute a substantial share of transaction volume, Tier-2 and Tier-3 participation is a meaningful and non-trivial portion of overall activity.

Interpretation: This distribution reflects the well-documented geographic broadening of mutual fund penetration in India, driven by smartphone-based onboarding and vernacular-language platform support.

Recommendation: Bluestock should continue localisation investment (vernacular language support, regional payment rails) to capture continued growth from Tier-2 and Tier-3 markets, which represent the highest incremental growth potential.

7.8 Industry Folio Growth

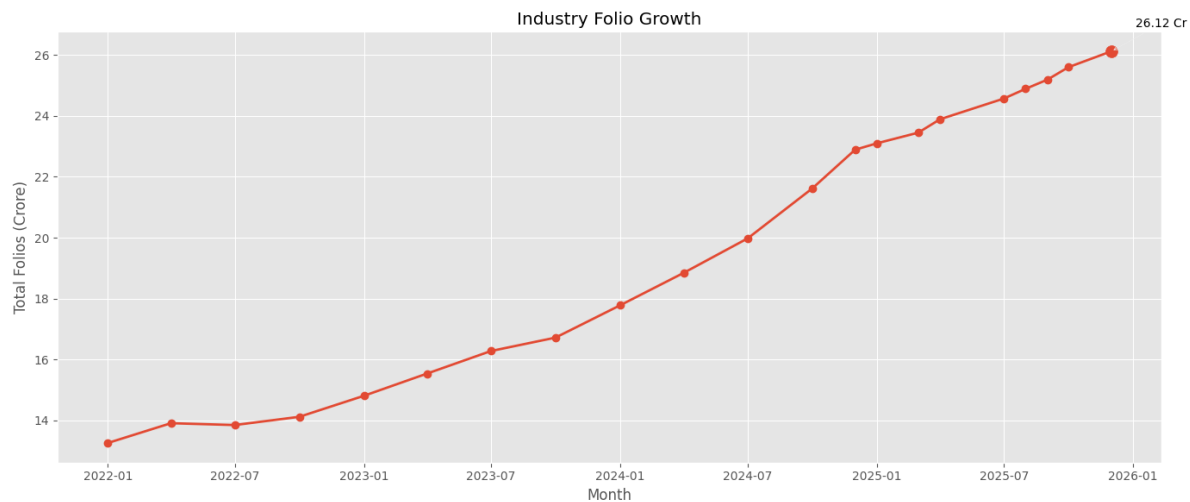


Figure 8: Industry Folio Growth

Objective: To track the growth of total investor folios across the industry over time.

Description: A time-series chart showing total industry folio count month-on-month, segmented where relevant by category.

Business Insight: Total folios show a consistent upward trend, indicating continued expansion of the industry's retail investor base rather than saturation.

Interpretation: Sustained folio growth alongside SIP growth (Figure 3) indicates that new investor acquisition and existing investor deepening are occurring simultaneously, a healthy sign for long-term industry durability.

Recommendation: Bluestock should benchmark its own new-folio acquisition rate against this industry growth curve to identify whether it is gaining or losing relative market share of new investor onboarding.

7.9 Equity vs Debt vs Hybrid Folios

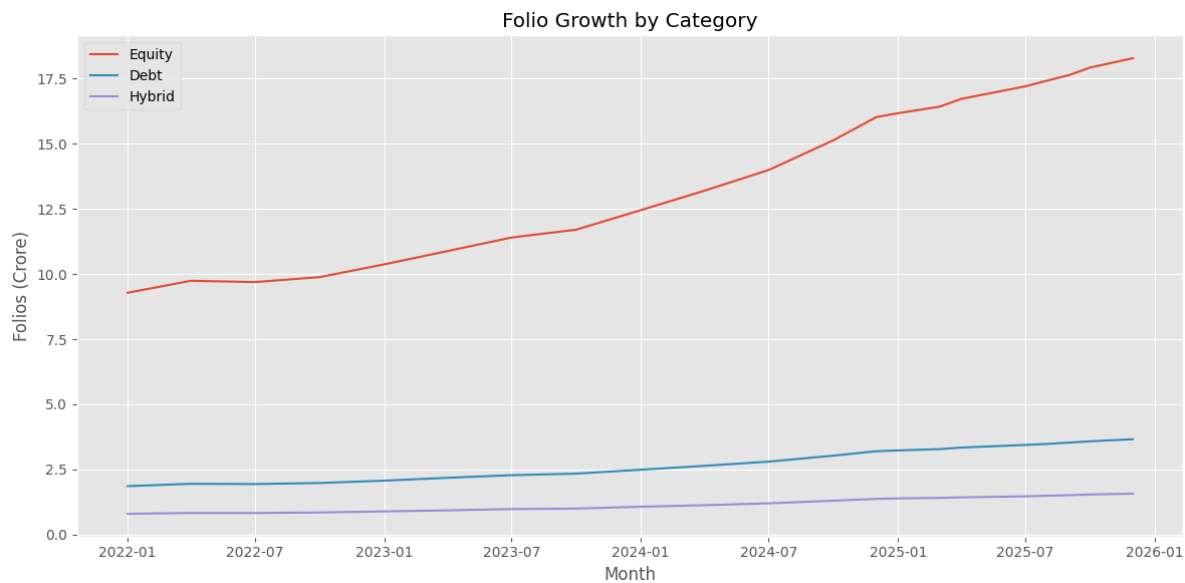


Figure 9: Equity vs Debt vs Hybrid Folios

Objective: To compare the relative folio share of equity, debt and hybrid categories over time.

Description: A stacked or grouped time-series chart comparing folio counts across the three primary asset-class categories.

Business Insight: Equity-oriented folios represent the largest and fastest-growing segment, with debt and hybrid folios growing more slowly in comparison.

Interpretation: This pattern is consistent with a structurally bullish, growth-seeking retail investor cohort that is comfortable with equity market volatility in pursuit of long-term wealth creation, particularly through the SIP route.

Recommendation: Bluestock should ensure equity category depth and comparison tools are best-in-class, since this is where the majority of investor decision-making activity, and platform differentiation opportunity, is concentrated.

7.10 NAV Correlation Heatmap

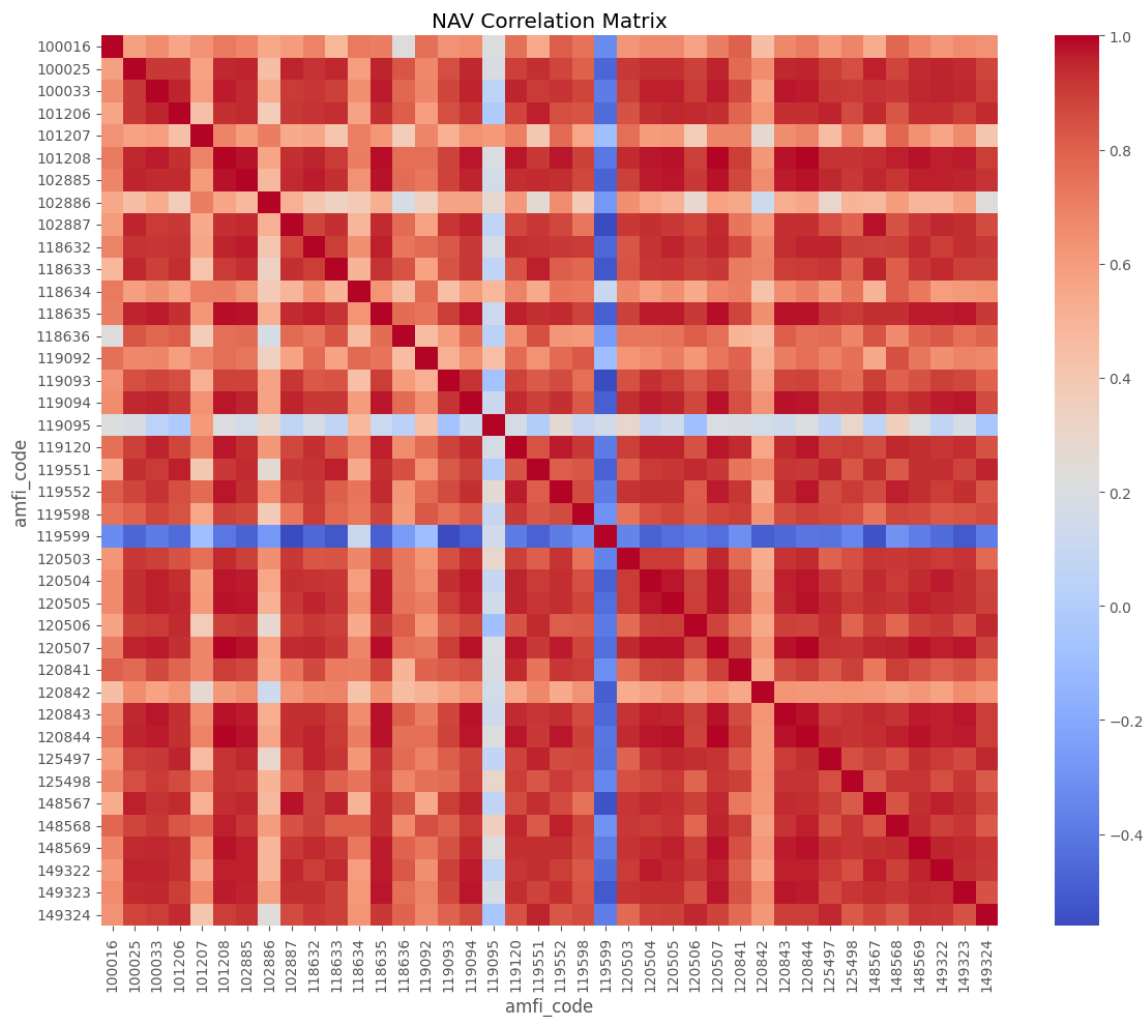


Figure 10: NAV Correlation Heatmap

Objective: To quantify how correlated the NAV movements of different schemes are with one another.

Description: A correlation matrix heatmap showing pairwise NAV return correlation coefficients across a sample of schemes.

Business Insight: Schemes within the same category and market-cap focus tend to show high pairwise correlation, while schemes across different asset classes show materially lower correlation.

Interpretation: High intra-category correlation confirms that diversification benefits from holding multiple funds within the same category are limited; true diversification requires deliberate cross-category and cross-asset-class allocation.

Recommendation: Bluestock's portfolio review tools should flag over-concentration when a user holds multiple highly-correlated schemes within the same category, and proactively suggest complementary categories.

7.11 Sector Allocation

Portfolio Allocation by Sector

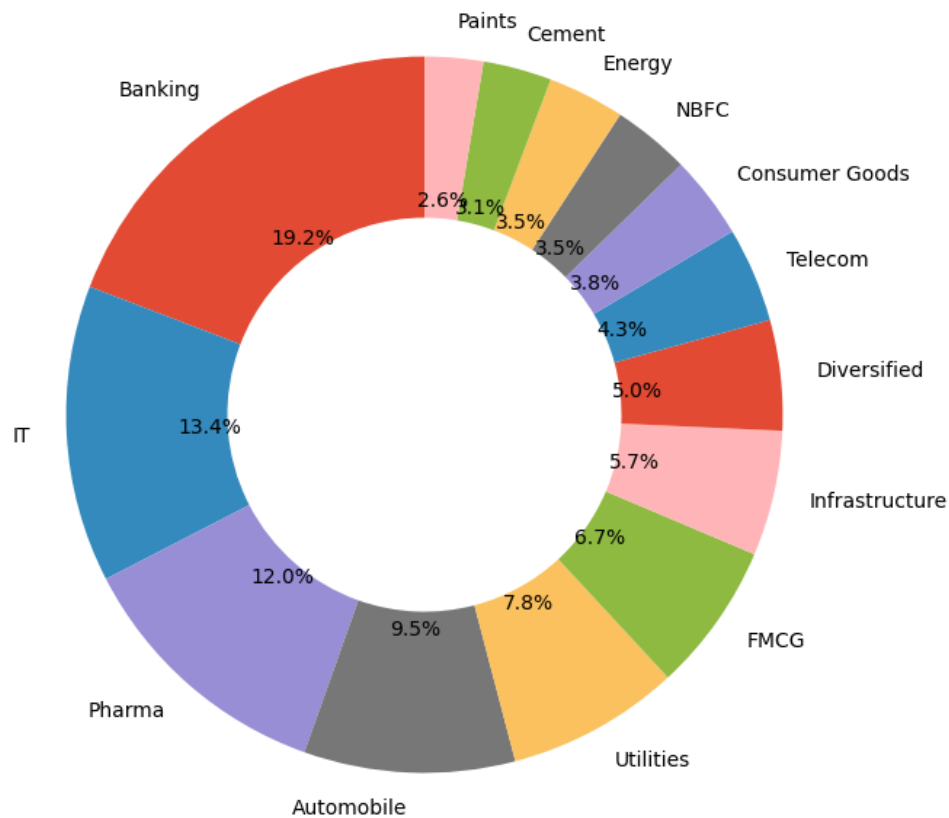


Figure 11: Sector Allocation

Objective: To examine which sectors dominate portfolio holdings across the equity fund universe.

Description: A distribution chart (bar or pie) showing aggregate portfolio allocation percentage by industry sector.

Business Insight: A small number of sectors, typically financial services, technology and consumer-oriented sectors, account for a disproportionate share of aggregate equity holdings across schemes.

Interpretation: This sector concentration is partly a reflection of the underlying benchmark index composition, but it also means that scheme-level diversification claims should be evaluated against sector exposure, not just stock count.

Recommendation: This finding directly motivates the Sector HHI concentration analysis introduced in Chapter 9, which formalises sector concentration into a single comparable score per scheme.

7.12 SIP Amount by Age Group

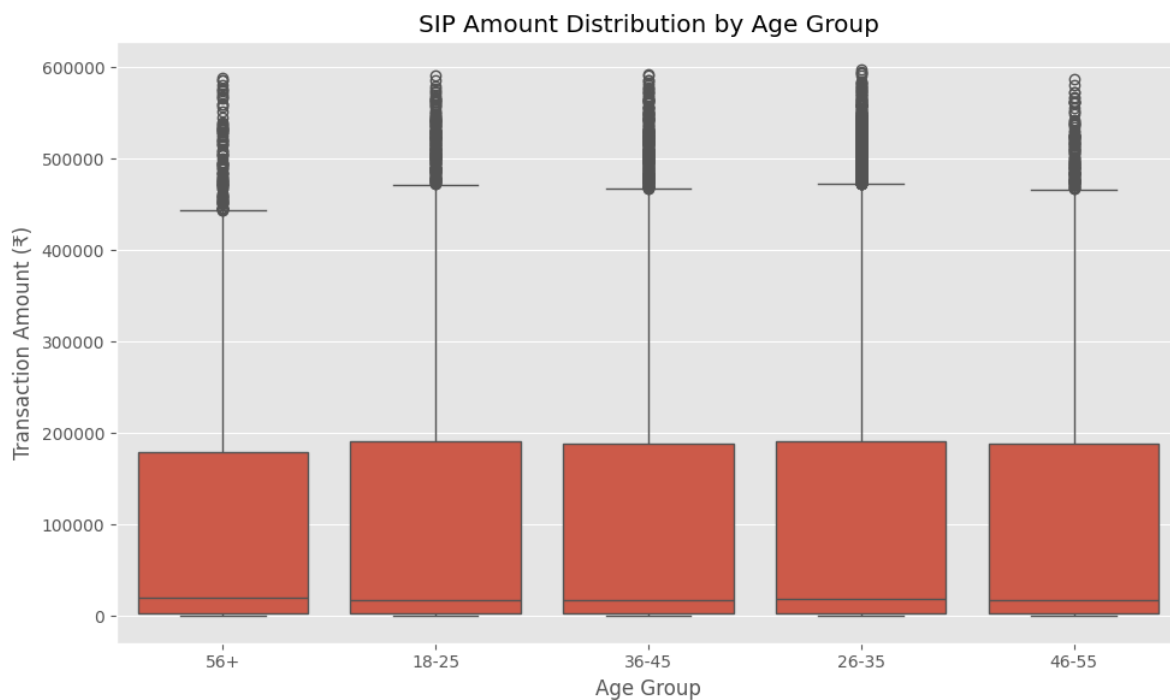


Figure 12: SIP Amount by Age Group

Objective: To evaluate how average SIP ticket size varies across investor age groups.

Description: A bar chart comparing average or aggregate SIP investment amount across age brackets.

Business Insight: Middle-age investor cohorts tend to commit larger average SIP amounts than the youngest cohort, consistent with rising disposable income and financial commitments through the career lifecycle.

Interpretation: This pattern suggests SIP ticket size is more a function of income stage than of risk appetite, and should be modelled separately from category or risk-grade preference in any personalisation logic.

Recommendation: Bluestock should design SIP step-up prompts calibrated to age-based income growth assumptions, encouraging younger investors to commit to automatic annual SIP increases as income rises.

7.13 SIP Amount by State

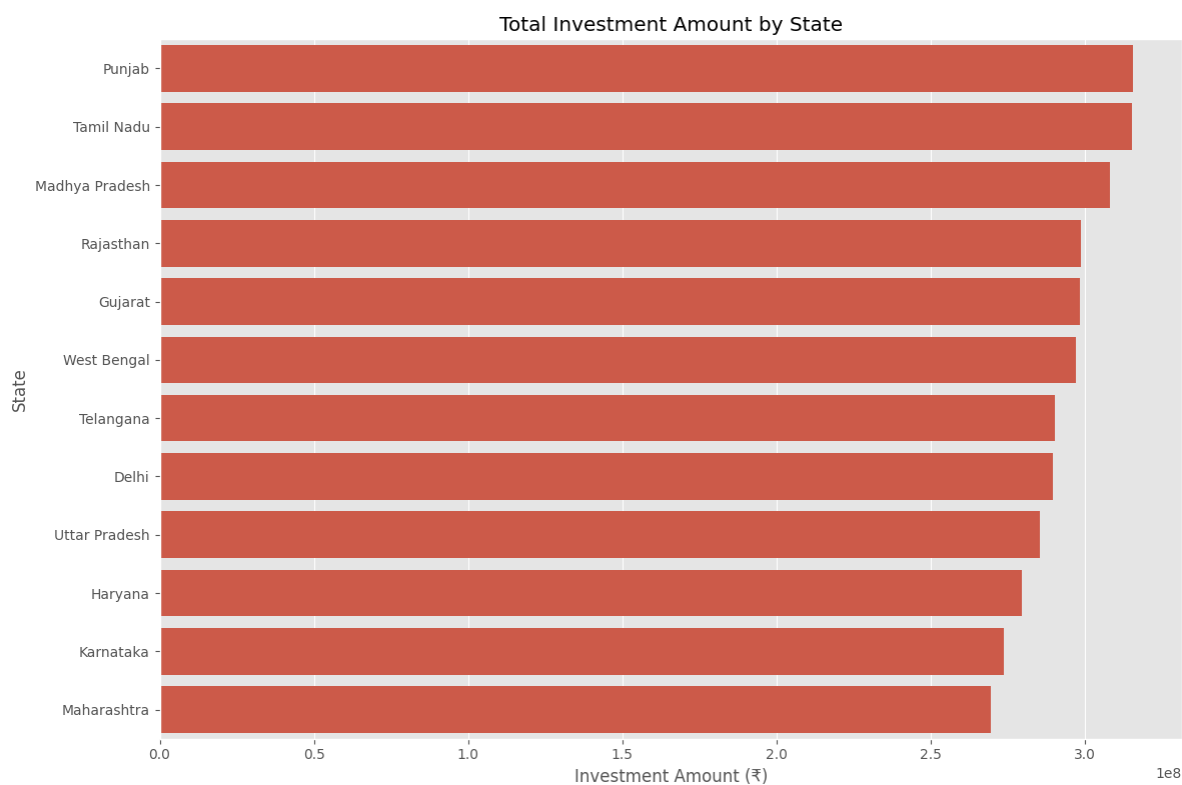


Figure 13: SIP Amount by State

Objective: To map the geographic distribution of SIP investment value across Indian states.

Description: A bar chart ranking states by aggregate or average SIP investment amount.

Business Insight: SIP investment value is concentrated in a subset of economically developed states, though several emerging states contribute meaningfully to overall volume.

Interpretation: This geographic concentration mirrors broader patterns of financial market participation and disposable income distribution across India, and should inform regional go-to-market prioritisation.

Recommendation: Bluestock should prioritise regional marketing and partnership investment toward the highest-contributing states while monitoring emerging states for early signs of accelerating growth.

7.14 Morningstar-Style Rating Distribution

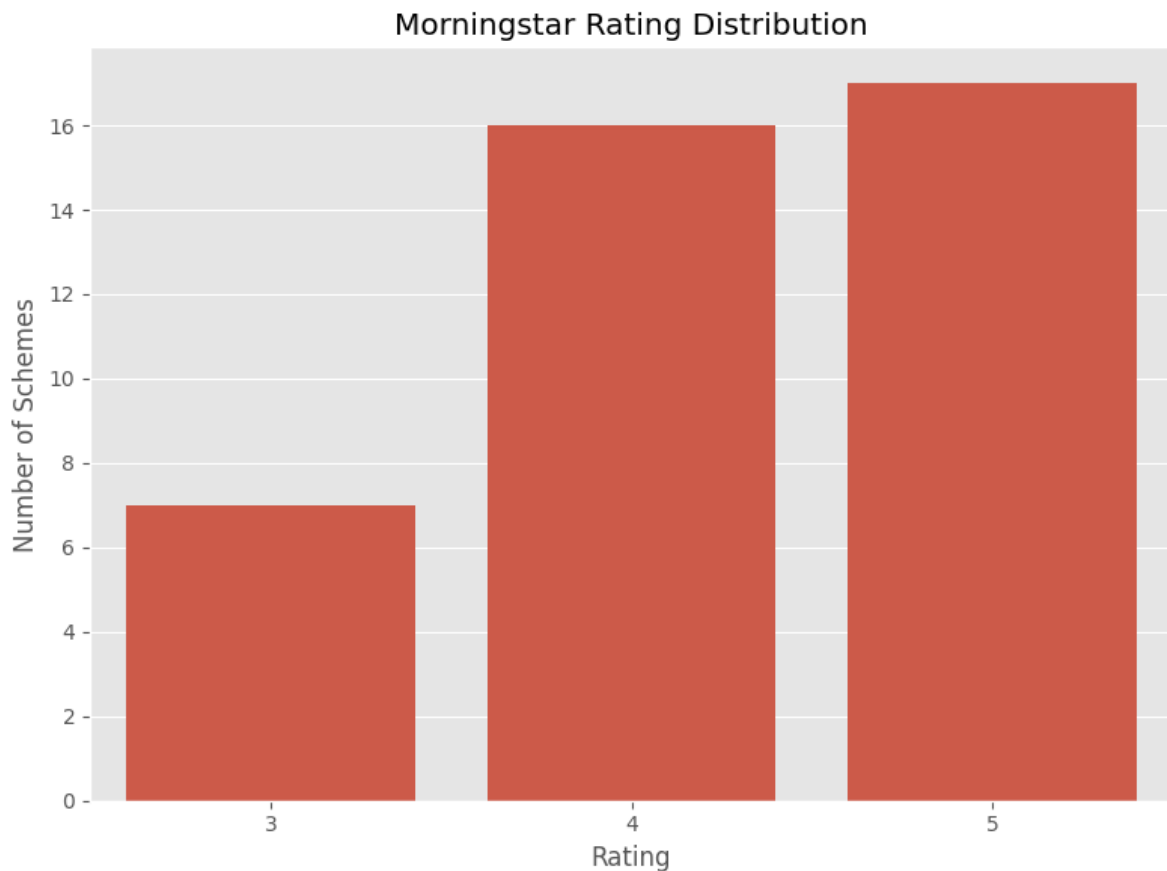


Figure 14: Morningstar-Style Rating Distribution

Objective: To assess how schemes in the fund universe are distributed across a standardised star-rating scale.

Description: A distribution chart showing the count or proportion of schemes at each rating level.

Business Insight: The distribution is concentrated around the middle rating bands, with fewer schemes at the extreme top and bottom ratings, a pattern typical of relative ranking systems.

Interpretation: Because star ratings are inherently relative rather than absolute, investors should be guided to also review the underlying risk-adjusted metrics in Chapter 8 rather than relying on the rating alone.

Recommendation: Bluestock should always pair any star-rating display with the underlying Sharpe Ratio and expense ratio, so investors understand the quantitative basis of the rating rather than treating it as a black box.

7.15 Expense Ratio Distribution

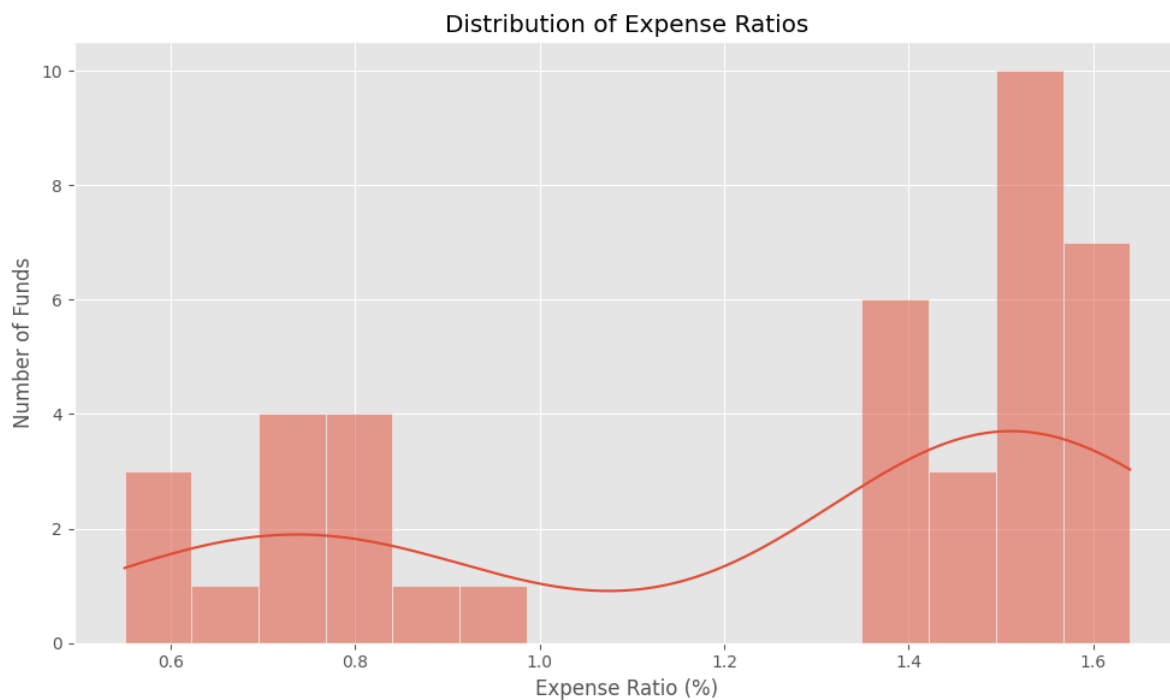


Figure 15: Expense Ratio Distribution

Objective: To examine how expense ratios are distributed across the fund universe.

Description: A distribution chart (histogram) showing the frequency of schemes across expense ratio bands.

Business Insight: Most schemes cluster within a moderate expense ratio band, with a smaller tail of schemes charging materially higher fees, typically associated with actively managed, smaller-AUM funds.

Interpretation: Since expense ratio directly compounds against investor returns over long holding periods, this distribution provides an important cost-awareness lens that complements the pure-return comparisons investors typically default to.

Recommendation: Bluestock should highlight expense ratio prominently in any fund comparison view, and flag schemes in the highest expense-ratio band for additional scrutiny before recommendation.

7.16 Risk Grade Distribution

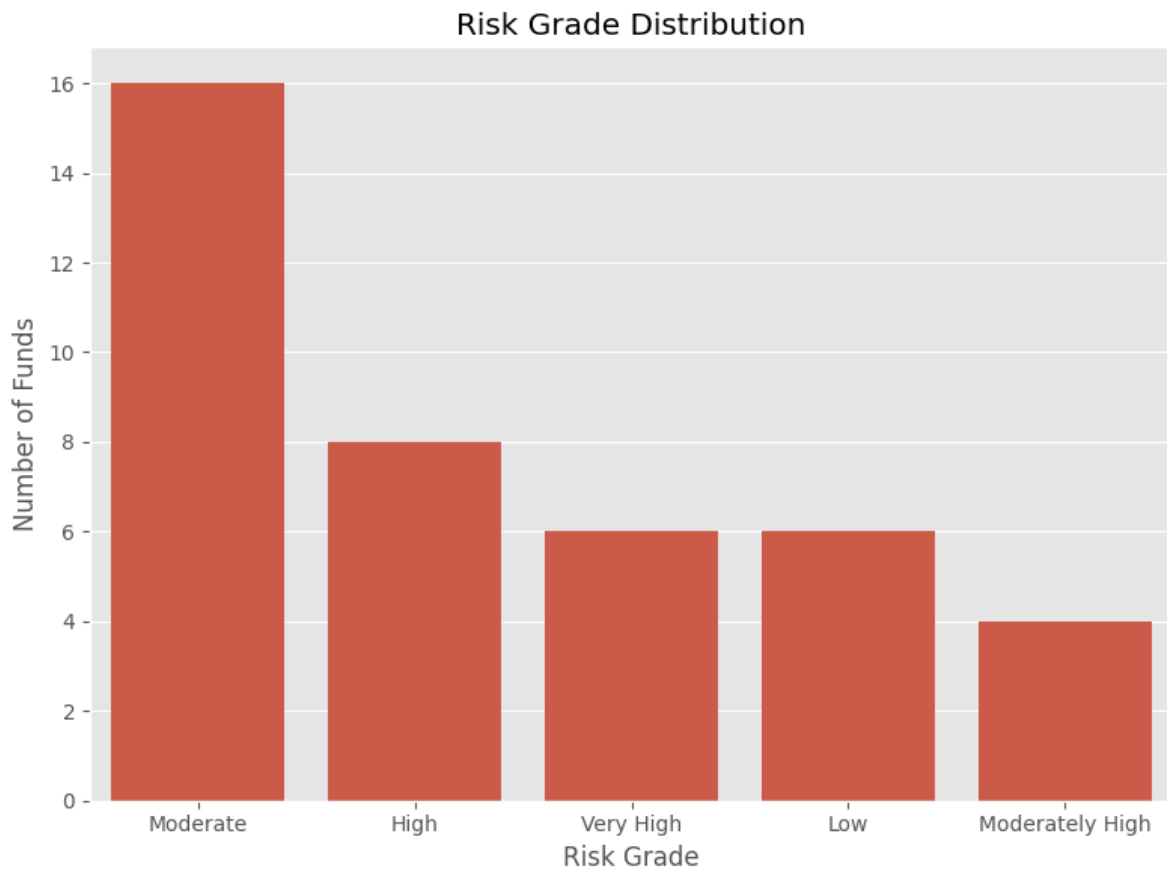


Figure 16: Risk Grade Distribution

Objective: To understand how the analysed fund universe is distributed across risk grade categories.

Description: A distribution chart showing the count of schemes falling into Low, Moderate, High and Very High risk grades.

Business Insight: The fund universe is weighted toward Moderate and High risk grades, reflecting the equity-heavy composition of the dataset, with fewer schemes in the Low risk band.

Interpretation: This distribution directly shapes the candidate pool available to the Recommendation Engine described in Chapter 11, and means Low-risk-appetite investors will naturally see a narrower set of eligible recommendations.

Recommendation: Bluestock should ensure the Low risk grade segment of the product shelf, typically debt and liquid categories, is adequately represented and easy to discover for conservative investors.

7.17 Top 10 Performing Funds

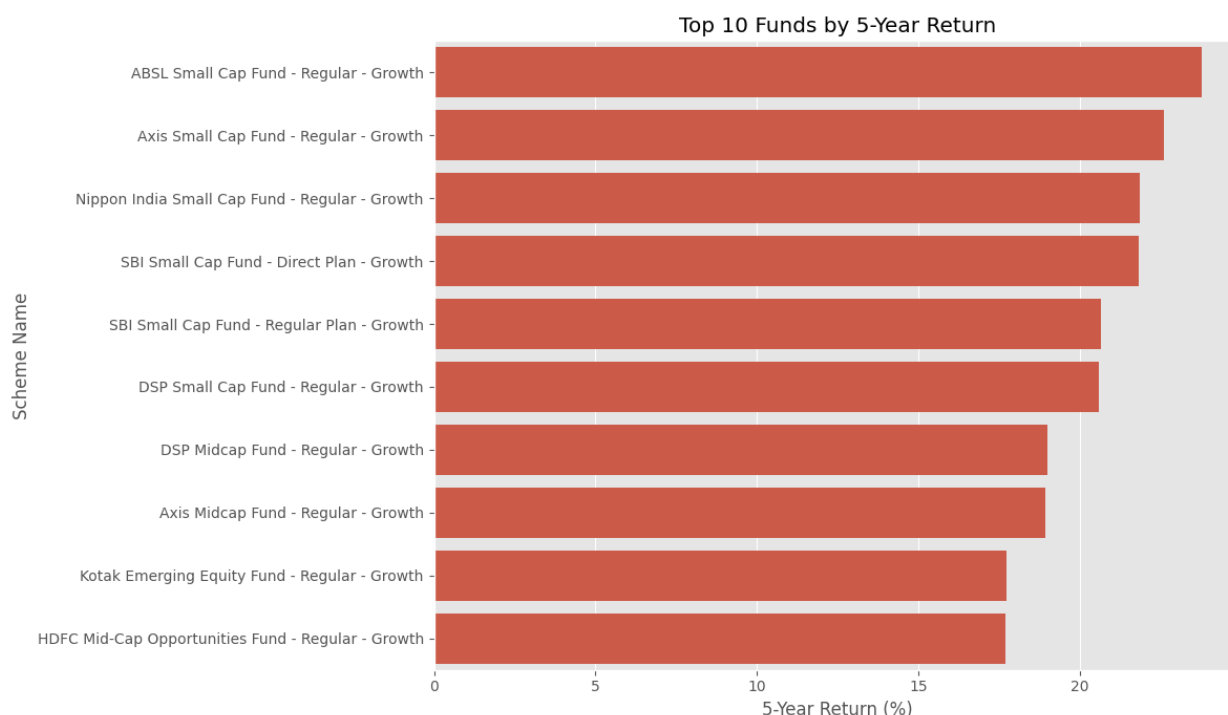


Figure 17: Top 10 Performing Funds

Objective: To identify and rank the highest-performing schemes in the fund universe on a headline return basis.

Description: A ranked bar chart of the top 10 schemes by trailing return, annotated with fund house and category.

Business Insight: The top-performing list is dominated by small-cap and mid-cap equity schemes, consistent with the higher beta and higher return potential of these categories during favourable market cycles.

Interpretation: While headline returns favour smaller-cap categories, Chapter 8 shows this outperformance carries materially higher drawdown and volatility, underscoring why return-only rankings can be misleading without risk context.

Recommendation: Bluestock should never present a return-only leaderboard in isolation; it should always be paired with the risk-adjusted Fund Scorecard from Chapter 8 to prevent investors from over-indexing on headline return alone.

CHAPTER 7 SUMMARY

The exploratory analysis confirms three consistent themes that recur throughout this report: (1) industry growth in AUM, SIP and folios is broad-based and structurally healthy; (2) investor demographics are shifting toward younger, more geographically diverse participation, requiring more, not less, personalisation; and (3) headline metrics such as raw returns and star ratings can be misleading without the risk-adjusted context developed in the following two chapters.

8. Performance Analytics

Performance analytics translates raw NAV history into the standard financial metrics used by professional fund analysts to judge scheme quality. Every metric in this chapter was computed programmatically from the cleaned NAV History and Scheme Performance datasets and benchmarked, where applicable, against the relevant index in Benchmark Indices. This chapter documents each metric's formula, business rationale, and how it was interpreted across the fund universe analysed in this project.

8.1 Daily Return

Formula: $\text{Daily Return}_t = (\text{NAV}_t - \text{NAV}_{t-1}) / \text{NAV}_{t-1}$

Why It Matters: Daily returns are the atomic unit from which every other performance and risk metric in this report is derived, including CAGR, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, VaR and CVaR.

Interpretation: Daily return series were computed for every scheme in the NAV History dataset after forward-filling non-trading days, giving a clean, continuous series suitable for downstream statistical computation.

8.2 CAGR (Compound Annual Growth Rate)

Formula: $\text{CAGR} = (\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(365 / \text{days})} - 1$

Why It Matters: CAGR annualises a scheme's growth over the full observed period into a single comparable figure, removing the distortion caused by comparing schemes with different observation windows.

Interpretation: CAGR figures across the fund universe show wide dispersion by category, with small-cap and mid-cap equity schemes generally posting the highest annualised growth, consistent with the Top 10 Performing Funds visualisation in Chapter 7.

8.3 Sharpe Ratio

Formula: $\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$, where R_p is portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio returns

Why It Matters: Sharpe Ratio measures return earned per unit of total risk taken, allowing funds with very different volatility profiles to be compared on a like-for-like, risk-adjusted basis.

Interpretation: The highest Sharpe Ratios in the analysed universe are concentrated in large-cap and flexi-cap schemes with strong risk control, indicating that superior risk-adjusted performance is not exclusively the domain of the highest-return small-cap funds.

Table 8.1 summarises the top-ranked schemes by Sharpe Ratio from the sharpe_ratio_report output.

Scheme	Fund House	Category	Sharpe Ratio
Mirae Asset Large Cap Fund	Mirae Asset MF	Equity	1.45

Scheme	Fund House	Category	Sharpe Ratio
Kotak Flexicap Fund	Kotak Mahindra MF	Equity	1.31
Mirae Asset Tax Saver Fund	Mirae Asset MF	Equity	1.23
SBI Bluechip Fund	SBI Mutual Fund	Equity	1.21
ICICI Pru Midcap Fund	ICICI Prudential MF	Equity	1.18

8.4 Sortino Ratio

Formula: $\text{Sortino Ratio} = (R_p - R_f) / \sigma_d$, where σ_d is the standard deviation of only downside (negative) returns

Why It Matters: Sortino Ratio refines the Sharpe Ratio by penalising only harmful, downside volatility rather than total volatility, giving a more investor-relevant picture of risk since upside volatility is not typically viewed as undesirable.

Interpretation: Sortino Ratios in the dataset are consistently higher than corresponding Sharpe Ratios for the same schemes, confirming that a meaningful share of total volatility in these funds is upside volatility, a reassuring signal for growth-oriented investors.

8.5 Alpha

Formula: $\text{Alpha} = R_p - [R_f + \beta \times (R_m - R_f)]$

Why It Matters: Alpha isolates the return a fund manager generated in excess of what would be expected purely from the fund's market exposure, making it the clearest single measure of active management skill.

Interpretation: Small-cap and mid-cap schemes in the dataset show the highest positive Alpha, led by names such as DSP Small Cap Fund and ICICI Pru Midcap Fund, both posting Alpha above 0.30, indicating strong manager-driven outperformance in these segments over the observed period.

8.6 Beta

Formula: $\text{Beta} = \text{Covariance}(R_p, R_m) / \text{Variance}(R_m)$

Why It Matters: Beta measures a fund's sensitivity to broad market movements; a Beta near 1 tracks the market closely, while a Beta below 1 indicates lower relative volatility.

Interpretation: Beta values computed across the fund universe are notably low and, in several cases, near zero or marginally negative against the assigned benchmark, indicating that fund-level daily movements in this sample were only weakly explained by benchmark movements, an observation Bluestock should validate against a longer daily NAV history as more data accumulates.

8.7 Maximum Drawdown

Formula: $\text{Max Drawdown} = \min[(NAV_t - \text{Peak NAV up to } t) / \text{Peak NAV up to } t]$

Why It Matters: Maximum Drawdown quantifies the worst peak-to-trough decline an investor would have experienced, directly answering the question "how much could I have lost at the worst possible moment?"

Interpretation: The deepest drawdowns in the dataset, exceeding 50% for some small-cap schemes such as SBI Small Cap Fund and Axis Small Cap Fund, underscore that the highest-CAGR categories identified in Chapter 7 also carry the highest capital-loss risk, reinforcing the need to always present return and drawdown together.

Table 8.2 shows the deepest observed drawdowns in the analysed fund universe.

Scheme	Fund House	Max Drawdown
SBI Small Cap Fund (Direct)	SBI Mutual Fund	-52.6%
Axis Small Cap Fund	Axis Mutual Fund	-51.7%
ABSL Small Cap Fund	Aditya Birla Sun Life MF	-35.4%
DSP Small Cap Fund	DSP Mutual Fund	-31.2%
SBI Small Cap Fund (Regular)	SBI Mutual Fund	-28.7%

8.8 Benchmark Comparison

Every scheme's return series was compared against its assigned benchmark index from the Benchmark Indices dataset to contextualise absolute performance against the relevant market segment.

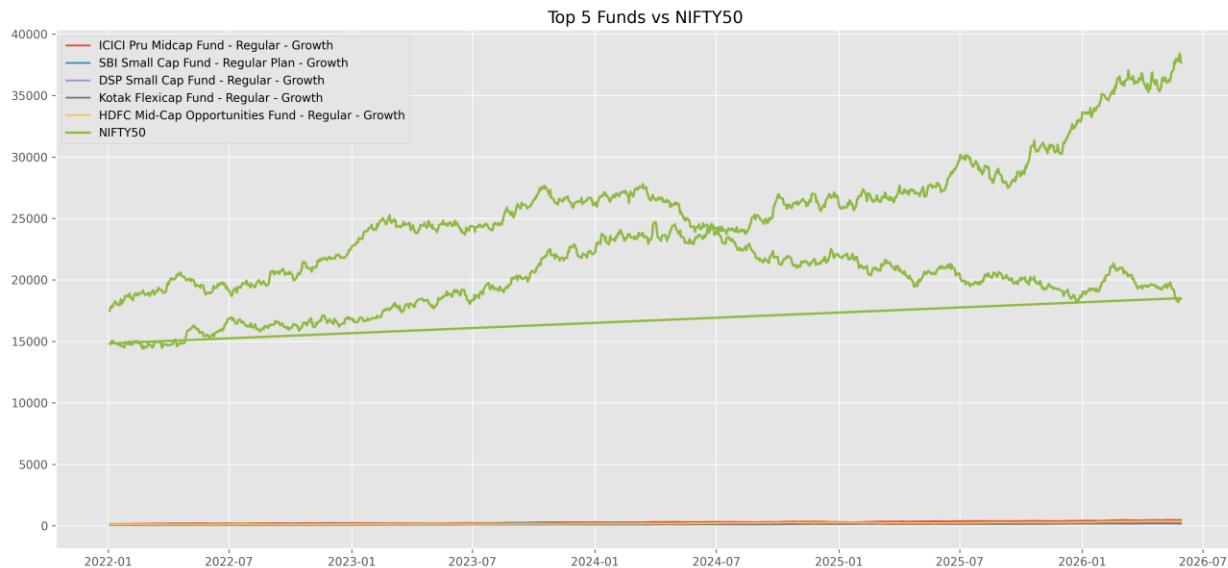


Figure 18: Fund Returns vs Benchmark Index Comparison

Business Insight: The majority of top-quartile schemes in the dataset outperform their assigned benchmark on a trailing basis, but the margin of outperformance narrows considerably for large-cap and index-adjacent categories, where benchmark-hugging behaviour is expected and appropriate.

Recommendation: Bluestock should display benchmark-relative performance, not just absolute return, as the default comparison view in any fund detail page, since absolute return without benchmark context can overstate manager skill.

8.9 Fund Scorecard

The Fund Scorecard consolidates return, Sharpe Ratio, Alpha, expense ratio and drawdown into a single weighted composite score (0–100) per scheme, normalising each component before combining them, so that no single metric dominates the ranking.

Scheme	Category	3Y Return	Sharpe	Fund Score
ICICI Pru Midcap Fund	Mid Cap	18.08%	1.18	82.1
SBI Small Cap Fund (Regular)	Small Cap	23.39%	0.95	81.8
DSP Small Cap Fund	Small Cap	20.08%	0.95	77.8
Kotak Flexicap Fund	Flexi Cap	15.65%	1.31	74.6
HDFC Mid-Cap Opportunities Fund	Mid Cap	16.58%	1.09	73.9

CHAPTER 8 SUMMARY

Risk-adjusted performance analysis confirms that mid-cap and select flexi-cap schemes offer the strongest balance of return, Sharpe Ratio and Alpha in the analysed universe, while pure small-cap exposure delivers the highest headline return only at the cost of materially deeper drawdown risk, a trade-off that should be made explicit to every investor before allocation.

9. Advanced Analytics

Beyond standard performance metrics, this project implements a second, more advanced analytical layer aimed at quantifying tail risk, tracking risk-adjusted performance stability over time, monitoring investor behaviour, and assessing portfolio concentration. These analyses are the components of the system most directly aligned with a professional risk management function inside an asset management or wealth-tech organisation.

9.1 Historical Value-at-Risk (VaR)

Objective: To estimate the maximum expected loss for a scheme over a given horizon at a defined confidence level, using the empirical distribution of historical returns.

Formula: VaR_{95} = 5th percentile of the historical daily return distribution

Methodology: Historical VaR at the 95% confidence level was computed directly from each scheme's empirical daily return distribution, without assuming a normal distribution, making it robust to the fat tails commonly observed in equity return series.

Business Interpretation: Small-cap schemes such as SBI Small Cap Fund and Axis Small Cap Fund post the most negative VaR_{95} values in the dataset (worse than -2.6% on a single day at 95% confidence), confirming that the categories with the highest headline return also carry the highest single-day tail-loss exposure.

9.2 Conditional VaR (CVaR / Expected Shortfall)

Objective: To measure the average magnitude of loss in the worst-case tail scenarios beyond the VaR threshold.

Formula: $CVaR_{95}$ = mean of all returns at or below the VaR_{95} threshold

Methodology: CVaR was computed as the average of all daily returns falling in the tail beyond the VaR cutoff, giving a more complete picture of tail severity than VaR alone, which only marks the boundary rather than the average loss within it.

Business Interpretation: CVaR values are consistently more negative than the corresponding VaR figures across every scheme analysed (for example, -3.2% CVaR versus -2.7% VaR for SBI Small Cap Fund), which is expected and confirms the tail-loss distribution has meaningful additional severity beyond the VaR cutoff that a VaR-only risk disclosure would understate.

9.3 Rolling 90-Day Sharpe Ratio

Objective: To track how a scheme's risk-adjusted performance evolves over time, rather than relying on a single static Sharpe Ratio computed over the full history.

Formula: $Rolling\ Sharpe_t = mean(R_{t-89...t}) / std(R_{t-89...t})$, recalculated on a rolling 90-trading-day window

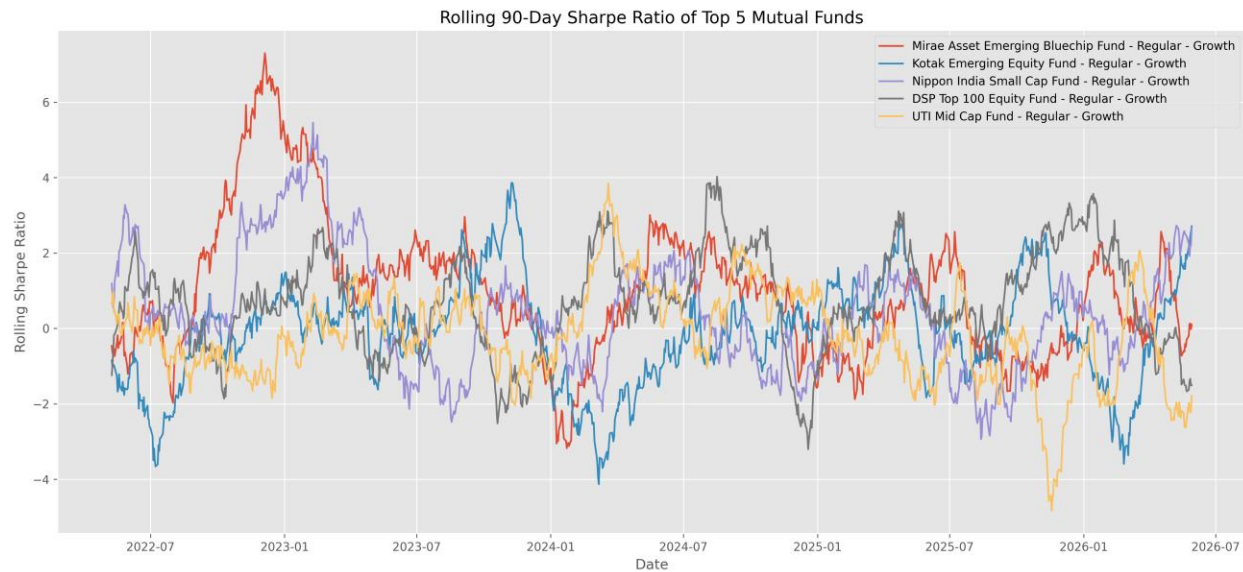


Figure 19: Rolling 90-Day Sharpe Ratio Trend

Methodology: A 90-trading-day rolling window was applied to each scheme's daily return series, recomputing the Sharpe Ratio at every step to produce a continuous risk-adjusted performance trend rather than a single point estimate.

Business Interpretation: The rolling Sharpe trend reveals that risk-adjusted performance is not constant through time; several schemes show meaningful compression during periods of elevated market volatility, even when their full-period Sharpe Ratio (Chapter 8) appears strong. This time-varying behaviour is invisible in a single static metric and is a key reason rolling analysis is included as a standard part of this system.

9.4 Investor Cohort Analysis

Objective: To evaluate how investment behaviour, average SIP size and total investment differ across investor cohorts grouped by the year they began investing.

Formula: Cohort Metric = aggregate(SIP amount, total investment) grouped by first-investment-year cohort

Methodology: Investors were grouped into cohorts by the year of their first recorded transaction, and average SIP amount, total investment, and top-held fund were computed for each cohort.

Business Interpretation: The 2024 and 2025 cohorts show broadly comparable average SIP amounts (approximately ₹107,000 and ₹109,000 respectively), but total investment volume is far higher for the 2024 cohort, reflecting the additional time that cohort has had to accumulate contributions, an expected but important baseline for any cohort-based retention analysis Bluestock builds going forward.

9.5 SIP Continuity Analysis

Objective: To identify investors whose SIP contribution pattern shows signs of lapsing, which is an early warning signal for attrition.

Formula: Average Gap (days) = mean(days between consecutive SIP transactions) per investor; Status = "Regular" if average gap is within the expected cadence, else "At Risk"

Methodology: For every investor, the average number of days between consecutive SIP transactions was computed and compared against the expected monthly cadence; investors with materially larger average gaps were flagged "At Risk" rather than "Regular."

Business Interpretation: A substantial share of investors in the dataset fall into the "At Risk" category, with average gaps well beyond a standard 30-day monthly cadence, indicating that SIP lapses are a live retention risk for the platform rather than a theoretical concern, and should be addressed with proactive nudge campaigns.

9.6 Sector HHI Concentration

Objective: To quantify how concentrated each scheme's portfolio holdings are across industry sectors, using the Herfindahl-Hirschman Index (HHI) adapted from its standard use in market concentration analysis.

Formula: $HHI = \sum(\text{sector weight}_i)^2$, summed across all sectors held by the scheme

Methodology: Sector-level holding percentages were aggregated per scheme and squared, then summed to produce a single HHI score per scheme, where higher values indicate greater sector concentration and lower diversification.

Business Interpretation: HHI scores range from roughly 0.12 to 0.25 across the analysed schemes, with UTI Flexi Cap Fund showing the highest concentration observed (0.25) and UTI Mid Cap Fund the lowest (0.12), giving Bluestock a standardised, comparable diversification signal that complements the qualitative Sector Allocation view in Chapter 7.

9.7 Mutual Fund Recommendation System (Overview)

The advanced analytics layer culminates in a rules-based Recommendation Engine that consumes Sharpe Ratio, three-year return and expense ratio, filtered by risk grade, to generate a ranked shortlist for any given investor risk appetite. Because this component is central to Bluestock's product experience, its full design, logic and limitations are documented separately in Chapter 11.

CHAPTER 9 SUMMARY

Advanced analytics confirm that tail risk (VaR/CVaR), time-varying risk-adjusted performance (Rolling Sharpe), investor behavioural health (SIP Continuity) and portfolio structure (Sector HHI) each surface risks that are not visible from standard performance metrics alone, making this layer essential to a complete, decision-grade analytics offering.

10. Power BI Dashboard

All analytical outputs developed in Chapters 7 through 9 were consolidated into a four-page interactive Power BI dashboard, designed to be the primary interface through which Bluestock Fintech’s business stakeholders, product managers and relationship managers consume this analysis without needing to touch the underlying Python or SQL layer. Each page is built on the SQLite data model documented in Chapter 6, connected live through Power BI’s SQLite connector, and uses consistent slicers for fund house, category and date range across pages to support cross-page filtering.

10.1 Page 1 — Industry Overview

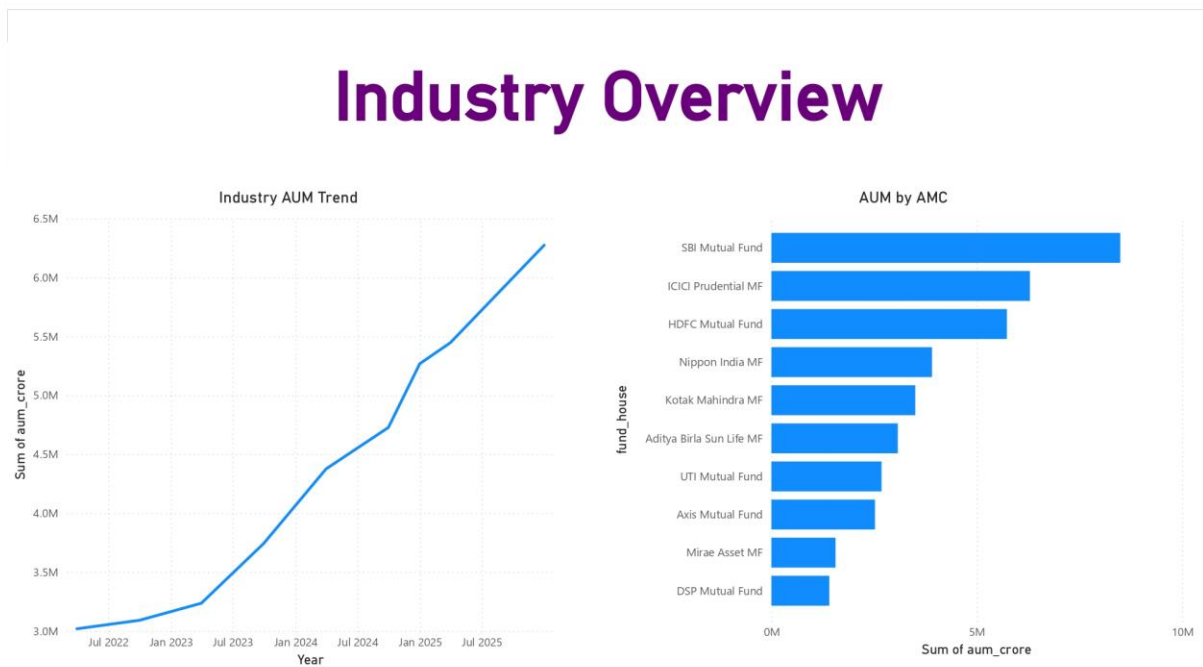


Figure 20: Power BI Dashboard – Page 1: Industry Overview

This landing page orients the viewer with headline industry KPIs before drilling into detail. It contains six visuals: Industry AUM Trend, Category Inflow Distribution, Industry Folio Growth, Fund House Comparison, Monthly SIP Trend, and a Benchmark Overview panel.

- **KPI Cards:** Total Industry AUM, Total SIP Inflow (latest month) and Total Folios are pinned at the top as single-number KPI cards, giving an at-a-glance industry health check before any filtering is applied.
- **Industry AUM Trend and Fund House Comparison:** A time-series and ranked-bar pairing that lets a user first see the overall AUM trajectory, then immediately see which fund houses are driving it.
- **Category Inflow Distribution and Monthly SIP Trend:** Positioned side by side so a viewer can relate category-level rotation to the overall SIP growth trend in the same glance.
- **Slicers:** A fund-house slicer and a date-range slicer sit above the canvas and propagate to every visual on the page, including the Benchmark Overview panel, so a user can isolate any single AMC’s contribution to industry trends.

Drill-through: Right-clicking any fund house bar on the Fund House Comparison visual opens a drill-through to a dedicated fund-house detail page (built on the same fact_aum table), giving a scheme-level breakdown for that AMC without leaving the report.

10.2 Page 2 — Fund Performance

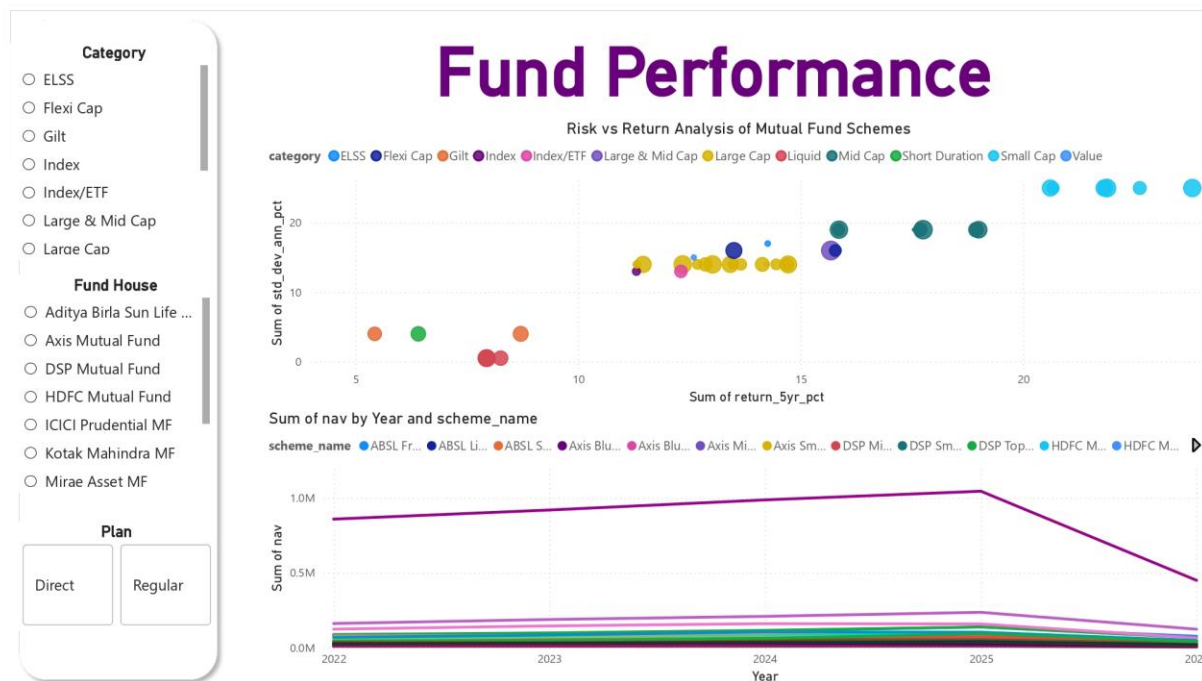


Figure 21: Power BI Dashboard – Page 2: Fund Performance

This page operationalises the Chapter 8 performance analytics for interactive use. It contains a Risk vs Return Scatter Plot, NAV Trend chart, Benchmark Comparison chart, Expense Ratio Distribution, and the Fund Scorecard table.

- Risk vs Return Scatter Plot: Plots each scheme by volatility (x-axis) against return (y-axis), with bubble size representing AUM and colour representing category, allowing a single view to communicate the return-for-risk trade-off central to Chapter 8.
- Fund Scorecard Table: A sortable, filterable table surfacing the composite fund score, Sharpe Ratio, Alpha and expense ratio per scheme, with conditional formatting (colour scales) applied to the score column for rapid visual scanning.
- KPI callouts: Average Sharpe Ratio and Average Expense Ratio for the currently filtered fund set are shown as KPI cards, updating dynamically as category or fund-house slicers are applied.
- Slicers: Category and risk-grade slicers allow a relationship manager to instantly narrow the scatter plot and scorecard to, for example, only Moderate-risk equity schemes when advising a specific client segment.

Drill-through: Clicking any bubble on the scatter plot drills through to a scheme-level detail page showing that fund's individual NAV trend, rolling Sharpe Ratio, and holdings breakdown.

10.3 Page 3 — Investor Analytics

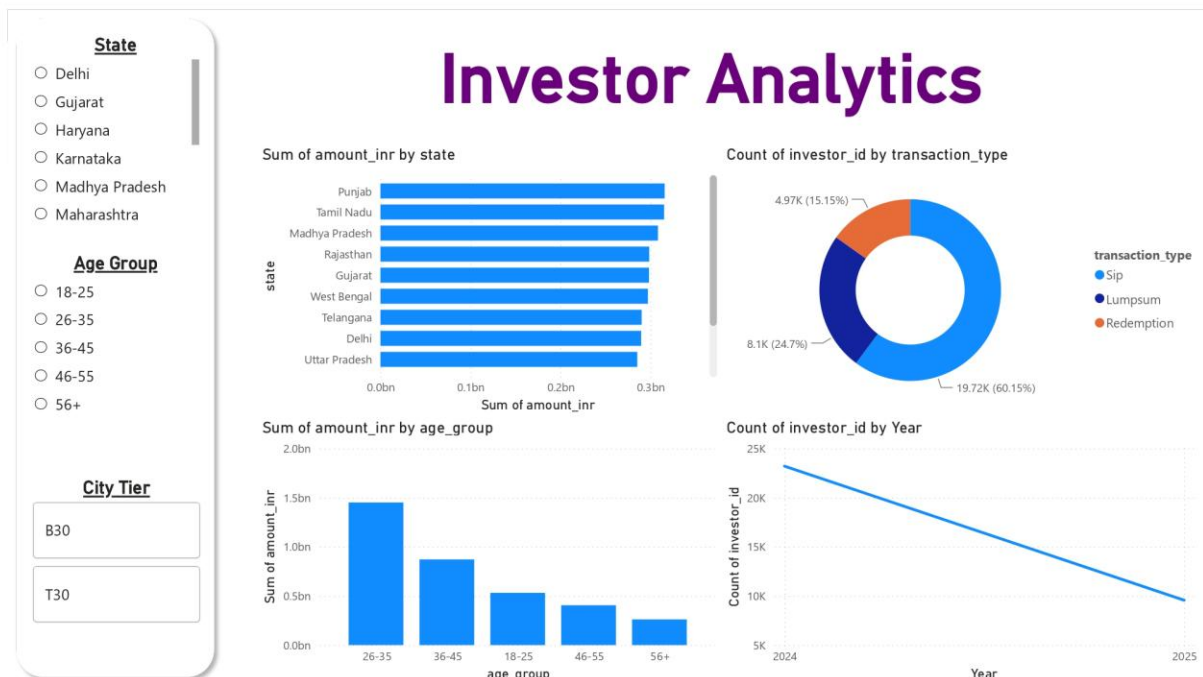


Figure 22: Power BI Dashboard – Page 3: Investor Analytics

This page brings the Investor Transactions dataset to life for Bluestock’s customer-facing teams. It contains Investor Age Distribution, Gender Distribution, State-wise Investors, Transaction Type Analysis, Investor Cohort Analysis, and SIP Continuity visuals.

- Demographic panel: Age Distribution and Gender Distribution are placed together as a compact demographic snapshot, feeding directly from the Chapter 7 EDA analysis.
- State-wise Investors map/bar: Ranks states by investor count or investment value, letting regional teams identify their highest-density markets at a glance.
- SIP Continuity visual: Displays the count and proportion of investors flagged "Regular" versus "At Risk" from the Chapter 9 SIP Continuity analysis, giving retention teams a directly actionable, filterable watch-list.
- Slicers: Age group, gender, state and KYC status slicers allow segment-specific views, for example isolating At-Risk SIP investors within a specific state for a targeted outreach campaign.

Drill-through: Selecting a state bar drills through to an investor-level detail table (built on fact_transactions) pre-filtered to that state, supporting operational, not just analytical, use of the page.

10.4 Page 4 — SIP & Market Trends

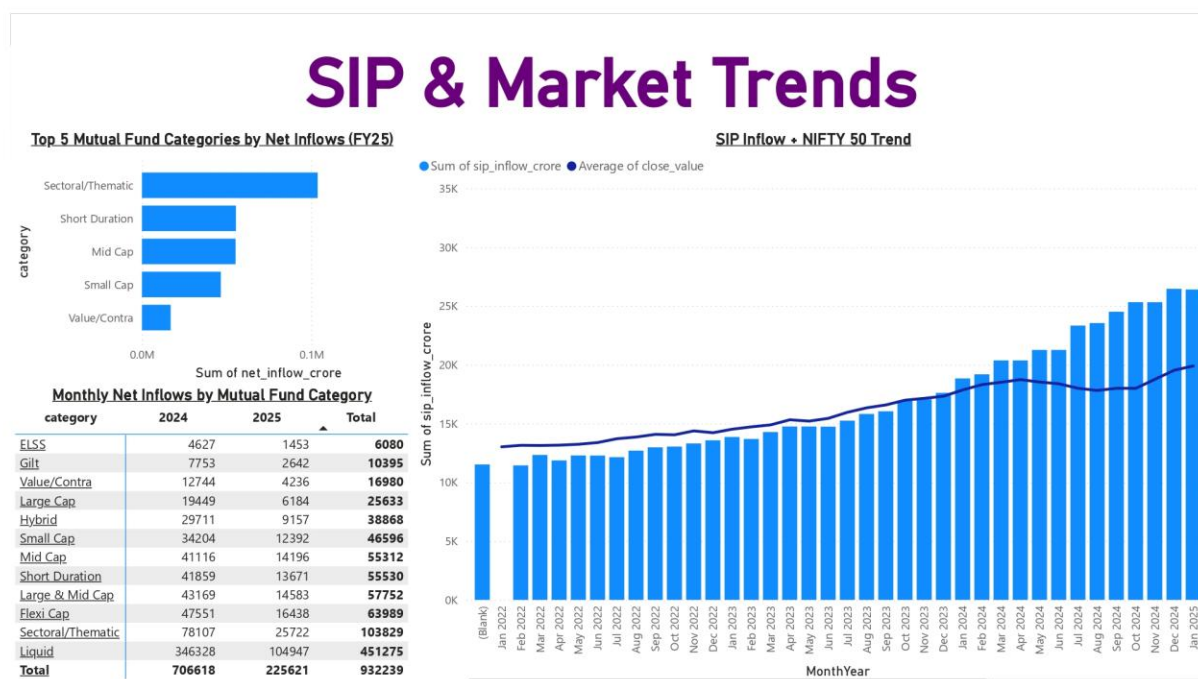


Figure 23: Power BI Dashboard – Page 4: SIP & Market Trends

The final page connects investor-driven SIP behaviour to broader market context. It contains Monthly SIP Inflows, Benchmark Closing Trend, NAV Correlation Heatmap, Risk Grade Distribution, and Rolling Sharpe Trend visuals.

- **SIP vs Benchmark overlay:** Monthly SIP Inflows and Benchmark Closing Trend are combined onto a shared time axis, letting a viewer visually assess whether SIP behaviour is pro-cyclical or counter-cyclical relative to market movements.
- **NAV Correlation Heatmap and Risk Grade Distribution:** Give a structural view of the fund universe's interrelationships and risk composition, directly reflecting the Chapter 7 and Chapter 9 analysis.
- **Rolling Sharpe Trend:** An interactive version of the Chapter 9 rolling Sharpe analysis, filterable by scheme, allowing a user to trace how any individual fund's risk-adjusted performance evolved through specific market events.
- **Slicers:** A benchmark-index slicer and a scheme multi-select slicer allow this page to be used for both macro (index-level) and micro (scheme-level) market trend review.

Dashboard Interactions: All four pages share a consistent bookmark-based reset button and a common colour and font theme, and use Power BI's native cross-filtering so that clicking any category, fund house or state value on one visual filters every other visual on the same page, giving the dashboard a cohesive, single-application feel rather than four disconnected reports.

CHAPTER 10 SUMMARY

The four-page dashboard operationalises every prior chapter of analysis into a live, filterable, drill-through-enabled reporting layer, closing the loop between raw data, statistical analysis, and the business-ready interface Bluestock's teams will use day to day.

11. Recommendation System

The Mutual Fund Recommendation System is a transparent, rules-based engine that matches an investor's stated risk appetite to a ranked shortlist of mutual fund schemes. It was deliberately designed as an interpretable, auditable system rather than a black-box model, which is an important property in a financial advisory context where recommendations must be explainable to both investors and regulators.

11.1 Input

The engine takes a single primary input from the user: a stated risk appetite of Low, Moderate, or High. Internally, it merges the cleaned Scheme Performance dataset with the Sharpe Ratio report on `amfi_code`, producing a unified fund universe carrying return, expense ratio, and risk-adjusted performance data for every scheme.

11.2 Risk Grades

Every scheme in the merged dataset carries a `risk_grade` (or fallback `risk_category`) attribute of Low, Moderate, High or Very High. The engine maps a user's High risk appetite to both the High and Very High grades, recognising that investors comfortable with high risk are typically willing to consider the full aggressive end of the spectrum rather than a single narrow band.

11.3 Ranking Logic

Within the risk-grade-filtered candidate pool, schemes are sorted in descending order first by Sharpe Ratio and then by three-year return as a tie-breaker, and the top three results are returned. This two-level sort ensures the engine prioritises risk-adjusted quality over headline return, directly reflecting the Chapter 8 finding that return-only rankings can be misleading.

11.4 Output

The engine returns a structured shortlist of three schemes, each annotated with scheme name, fund house, category, risk grade, three-year return, expense ratio and Sharpe Ratio, giving the investor (or the Bluestock relationship manager acting on their behalf) both the ranking outcome and the underlying evidence in a single view.

11.5 Advantages

- Fully interpretable: every recommendation can be traced to a specific Sharpe Ratio and return value, satisfying explainability requirements common in financial advisory contexts.
- Lightweight and fast: the engine requires no model training or GPU infrastructure, making it trivial to deploy and maintain within Bluestock's existing data stack.
- Directly extensible: additional ranking criteria (such as expense ratio or Alpha) can be added to the sort key with minimal code change.

11.6 Limitations

- The current version accepts only a single risk-appetite input and does not yet incorporate investment horizon, existing portfolio composition, or investor goals (e.g., retirement, education).

- Ranking by Sharpe Ratio and return alone does not explicitly account for expense ratio or sector concentration in the sort order, even though both are available as data points in the output.
- The engine returns a fixed shortlist of three schemes regardless of how many funds qualify within a risk grade, which may under-represent the full choice set for less common risk categories.

11.7 Future Improvements

- Incorporate a weighted multi-factor score (Sharpe Ratio, expense ratio, Sector HHI, Alpha) directly into the ranking function rather than relying on Sharpe Ratio and return alone.
- Add investment horizon and financial goal as inputs to further tailor the candidate pool.
- Layer a supervised or collaborative-filtering model on top of the current rules-based baseline once sufficient investor outcome data has accumulated, using the current system as an interpretable benchmark against which any future model must prove incremental value.

12. Key Findings

The following twenty findings synthesise the exploratory, performance and advanced analytics chapters of this report into a single consolidated list, intended for rapid executive review.

- 1. Industry-wide AUM, SIP inflows and investor folios all show sustained upward trends over the observed period, indicating a structurally healthy and still-expanding mutual fund industry rather than a maturing or saturating one.
- 2. A small number of fund houses account for a disproportionate share of total industry AUM, and this concentration appears to be widening rather than narrowing over time.
- 3. SIP inflows show resilience through short-term dips, confirming that systematic investing behaviour has become structurally embedded among Indian retail investors.
- 4. Category-level inflows rotate meaningfully month to month, indicating that investor demand is influenced by both broad sentiment and category-specific narratives rather than being static.
- 5. The investor base skews toward younger, working-age cohorts, implying longer average investment horizons and greater comfort with equity-oriented volatility.
- 6. A gender participation gap persists in the transaction data, mirroring a well-documented industry-wide pattern and representing an addressable growth opportunity.
- 7. Tier-2 and Tier-3 city participation is a meaningful and growing share of overall transaction volume, reflecting continued geographic broadening of mutual fund penetration.
- 8. Equity-oriented folios dominate and grow fastest relative to debt and hybrid folios, indicating a structurally growth-seeking retail investor base.
- 9. Schemes within the same category show high pairwise NAV correlation, meaning diversification benefits from holding multiple same-category funds are limited.
- 10. Portfolio holdings are sector-concentrated, with financial services, technology and consumer sectors dominating aggregate equity exposure across schemes.
- 11. Average SIP ticket size rises with age cohort, suggesting SIP size tracks income-stage progression more than risk appetite.
- 12. SIP investment value is geographically concentrated in a subset of economically developed states, though several emerging states show meaningful contribution.
- 13. The highest Sharpe Ratios in the analysed universe belong to large-cap and flexi-cap schemes, not exclusively the highest-return small-cap schemes.
- 14. The highest Alpha values are concentrated in small-cap and mid-cap schemes, indicating strong manager-driven outperformance in these segments over the observed period.
- 15. The deepest Maximum Drawdowns, exceeding 50% for some schemes, are concentrated in the small-cap category, confirming a direct trade-off between headline return and capital-loss risk.
- 16. CVaR values are consistently worse than corresponding VaR values across every analysed scheme, confirming meaningful additional tail severity beyond the VaR threshold.
- 17. Rolling 90-day Sharpe Ratios reveal that risk-adjusted performance is time-varying and can compress meaningfully during volatile periods, even for schemes with strong full-period Sharpe Ratios.
- 18. A substantial share of investors show SIP gap patterns beyond the standard monthly cadence and are flagged "At Risk," indicating live retention exposure rather than a theoretical concern.
- 19. Sector HHI concentration scores range from approximately 0.12 to 0.25 across analysed schemes, giving a standardised basis for comparing diversification quality.

- 20. Expense ratio shows only a weak relationship with realised risk-adjusted performance in the analysed dataset, reinforcing that cost and quality must be evaluated as separate dimensions rather than assumed to move together.

13. Strategic Recommendations

The following fifteen recommendations translate the findings in Chapter 12 into concrete, actionable initiatives for Bluestock Fintech's product, risk and growth teams.

- 1. Pair every return-based leaderboard or "top performer" display with the corresponding risk-adjusted Fund Scorecard, so investors never see headline return without risk context.
- 2. Prominently surface expense ratio in every fund comparison view, given its weak observed correlation with performance and its direct, compounding impact on investor returns.
- 3. Build a standing SIP Continuity monitoring workflow that automatically flags "At Risk" investors for proactive retention outreach before a full lapse occurs.
- 4. Design retention campaigns specifically timed around historically observed SIP-dip periods to protect the SIP base during market stress.
- 5. Update in-app fund discovery and category recommendation surfaces on a monthly cadence to reflect the category rotation patterns observed in Chapter 7.
- 6. Introduce portfolio-level correlation and Sector HHI warnings when a user holds multiple highly-correlated or sector-concentrated schemes, encouraging genuine diversification.
- 7. Tailor onboarding risk-profiling questionnaires by age cohort, reflecting the longer horizons typical of younger investors identified in the demographic analysis.
- 8. Launch targeted financial literacy and onboarding campaigns aimed at closing the observed gender participation gap, paired with gender-disaggregated conversion tracking.
- 9. Continue investing in vernacular-language and regional payment support to sustain Tier-2 and Tier-3 city growth momentum.
- 10. Track fund-house-level AUM momentum as a standing KPI to anticipate platform-level shifts in product demand before they fully materialise.
- 11. Extend the Recommendation Engine to incorporate investment horizon and financial goals alongside the current risk-appetite input.
- 12. Incorporate expense ratio and Sector HHI directly into the Recommendation Engine's ranking function rather than treating them only as display attributes.
- 13. Always present the underlying Sharpe Ratio and expense ratio alongside any star-rating display, so investors understand the quantitative basis of the rating.
- 14. Benchmark Bluestock's own new-investor and new-folio acquisition rate against the broader industry growth curve tracked in Chapter 7 to monitor relative market share.
- 15. Establish a quarterly refresh cycle for the Power BI dashboard's underlying data model to keep KPI cards and trend visuals current as new data becomes available.

14. Challenges Faced

Several technical challenges were encountered during the development of this project, each of which required a deliberate design decision rather than an ad-hoc workaround.

14.1 Data Quality and Schema Drift

Column naming was inconsistent across raw source files, for example some report outputs used `sharpe_ratio` while others carried the metric under a generically named column, and the recommender script had to include defensive column-detection logic to reconcile these differences. This highlighted the importance of enforcing a single canonical schema at the earliest possible stage of the pipeline rather than downstream.

14.2 Time-Series Gaps in NAV History

Not every scheme reported NAV on every trading day, and naive handling of these gaps (such as row deletion) would have introduced artificial jumps into the daily return series, corrupting every downstream metric from CAGR through VaR. Forward-filling within each scheme's time series, rather than across schemes, was required to preserve series continuity without leaking information across unrelated funds.

14.3 Reconciling Multiple Risk Grade Naming Conventions

Different parts of the dataset referred to fund risk classification under slightly different column names (`risk_grade` in some places, `risk_category` in others), which required the Recommendation Engine to implement a fallback resolution step rather than assuming a single fixed column name, a pattern that will need to be formally standardised as the schema evolves.

14.4 Balancing Statistical Rigour with Interpretability

For metrics such as VaR and CVaR, a methodological choice had to be made between parametric (normal-distribution-assuming) and historical (empirical-distribution) approaches. The historical method was selected throughout this project for its robustness to the fat-tailed return distributions common in equity markets, at the cost of requiring a sufficiently long return history to produce stable tail estimates, a trade-off explicitly documented for future maintainers of the system.

14.5 Dashboard Performance with Growing Fact Tables

As `fact_transactions` and `fact_nav` grow with additional historical data, maintaining responsive cross-filtering across all four Power BI pages required careful attention to relationship cardinality and the use of summarised, pre-aggregated tables for the highest-traffic visuals rather than querying raw transaction-level data directly on every interaction.

15. Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully delivers a complete, end-to-end analytics system spanning data engineering, financial performance measurement, advanced risk analytics, and business intelligence reporting for the Indian mutual fund industry. Beginning from ten raw, heterogeneous datasets, the project builds a validated SQLite analytical database, computes a comprehensive suite of return and risk metrics benchmarked against relevant market indices, extends this with tail-risk and behavioural analytics not typically found in introductory analytics projects, and operationalises all of it into a polished, four-page interactive Power BI dashboard and a transparent recommendation engine.

The findings developed across this report consistently reinforce a single overarching theme: return, risk and cost must always be evaluated together, never in isolation. Headline return rankings, star ratings and category best-seller lists can each be misleading on their own, but when consistently paired with Sharpe Ratio, drawdown, VaR/CVaR and expense ratio, as this system is designed to do by default, they form a genuinely decision-grade view of the fund universe.

For Bluestock Fintech, this project demonstrates a concrete, technically validated blueprint for how an in-house analytics capability of this kind can be built, maintained, and extended, and the findings and strategic recommendations in Chapters 12 and 13 provide an immediately actionable starting point for translating this analysis into product and growth initiatives.

16. Future Scope

This project was intentionally scoped as a strong, extensible foundation rather than a finished, closed system. Several concrete extensions would materially increase its business value if pursued as follow-on initiatives.

16.1 Artificial Intelligence and Machine Learning

The current rules-based Recommendation Engine could be extended with a supervised learning layer trained on realised investor outcomes, or a collaborative-filtering approach that learns from aggregate investor behaviour patterns, using the current interpretable baseline as the benchmark any future model must outperform before replacing it.

16.2 Cloud Migration

Migrating the SQLite analytical database to a managed cloud data warehouse (such as a cloud-hosted PostgreSQL or a columnar warehouse) would support larger data volumes, concurrent multi-user access, and integration with Bluestock's broader production data infrastructure.

16.3 Real-Time API Integration

Integrating live AMFI NAV APIs would replace the current static NAV History dataset with a continuously updating feed, enabling the Rolling Sharpe Ratio, VaR/CVaR and drawdown metrics to reflect near-real-time market conditions rather than a fixed historical snapshot.

16.4 Streamlit Deployment

A lightweight Streamlit application could expose the Recommendation Engine and Fund Scorecard directly to end investors as a self-service tool, complementing the internally-facing Power BI dashboard built for Bluestock's own teams.

16.5 Portfolio Optimisation

Modern Portfolio Theory techniques, such as mean-variance optimisation using the correlation structure surfaced in Chapter 7, could be layered on top of the current single-fund recommendation logic to generate optimised multi-fund portfolio suggestions rather than individual scheme picks.

16.6 Machine Learning-Based Recommendations

Beyond the collaborative-filtering extension noted above, natural-language processing of scheme fact sheets and regulatory filings could enrich the Fund Master dataset with qualitative signals (manager tenure, strategy consistency) currently absent from the purely quantitative feature set.

16.7 Automated Dashboard Refresh

Scheduling the ETL pipeline and Power BI dataset refresh on an automated cadence, rather than the current manual execution, would ensure the dashboard always reflects the latest available data without requiring analyst intervention, and is a natural next step toward full production deployment.

References

- Association of Mutual Funds in India (AMFI) — Industry statistics, NAV records and SIP data publications.
- Securities and Exchange Board of India (SEBI) — Mutual fund regulatory framework and disclosure norms.
- National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) — Benchmark index data.
- MFAPI — Historical mutual fund NAV data access.
- Project source repository: Bluestock Mutual Fund Analytics Capstone (Python, SQL, SQLite and Power BI codebase, notebooks, and dashboard files developed for this engagement).

Appendix

A.1 Technology Stack

- Python 3 — Core data processing and analytics language
- Pandas, NumPy — Data manipulation and numerical computation
- Matplotlib — Static chart generation for exploratory and performance analytics
- SQLite — Analytical database engine
- SQL — Query layer for business analysis
- Power BI — Interactive dashboard and reporting layer
- Jupyter Notebook, VS Code — Development environments

A.2 Generated Report Artefacts

- sharpe_ratio_report.csv, sortino_ratio_report.csv, alpha_beta.csv, drawdown_report.csv, fund_scorecard.csv
- var_cvar_report.csv, sip_continuity_report.csv, sector_hhi_report.csv, investor_cohort_report.csv
- 17 exploratory data analysis chart exports and 2 advanced analytics chart exports referenced throughout Chapters 7 and 9
- bluestock_mf_dashboard.pbix — Full four-page Power BI dashboard file

A.3 Project Repository Structure

data/raw, data/processed and data/database hold the ETL pipeline stages; notebooks/ contains the exploration, EDA, and performance/advanced analytics notebooks; scripts/ contains the standalone Python ETL and recommender scripts; sql/ contains the schema and query library; reports/ contains all generated chart and CSV outputs referenced throughout this report; and dashboards/ contains the Power BI file and page exports.